Forecasting with Kernel Affine Hull Machines: Representation Refinement and Risk Certification

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Abstract

Kernel Affine Hull Machines (KAHMs) represent data through local geometric reconstruction. We use their soft associations to forecast observed signals and assess accuracy against a specified error tolerance. Retaining dictionary blocks under a bound on the sum of block Frobenius norms preserves earlier predictors and prevents duplication from enlarging the feasible class. We derive a population risk lower bound from pairs whose target separation exceeds that allowed by the decoder's Lipschitz budget. This pairwise obstruction is exact for a distribution supported on two points and dominates a conditional root-variance bound under the same pairing law. Calibration on independent episodes combines this obstruction with fitted-risk upper bounds and paired improvement comparisons, allowing unresolved accuracy decisions. The theory extends to other fixed representations. Experiments cover a known-risk benchmark, six synthetic dynamics, and writer-separated pen trajectories. Matched controls show KAHM gains on nonlinear autoregression and pen trajectories, while affine-hull alternatives perform better on oscillators. Preprocessing produces both gains and regressions. Follow-ups on the same writer cohort are descriptive. The guarantees concern expected squared error under a specified episode and forecast-origin law; few independent evaluation units can leave the bounds uninformative.

Keywords: Kernel Affine Hull Machines, time-series forecasting, soft state abstraction, representation refinement, prediction risk certification

1 Introduction

Kernel Affine Hull Machines (KAHMs) use geometric reconstruction models in reproducing kernel Hilbert spaces to describe data (Kumar et al., 2024, 2025). For a time-series history, each local model measures reconstruction quality relative to a learned reference group. Normalizing these measurements gives a vector of soft associations. A decoder, also called a prediction head, maps that vector to future observed values. The central question is whether the representation retains the distinctions needed for forecasting. Similar associations can conceal different futures, and fitting a larger dictionary need not improve predictions.

Neither good reconstruction nor accurate prediction of the feature coordinates settles this question. The coordinate target changes with the dictionary, and an almost constant representation can be easy to predict while losing useful information. We therefore use the future observed signal as the common target throughout learning, refinement, and evaluation. For each forecast horizon h and requested root-mean-square error (RMSE) tolerance $\epsilon_h \geq 0$, we ask two questions: does a fitted forecast meet the tolerance, and can *any* decoder with a prescribed sensitivity meet it using this representation? Both questions concern the same distribution of new trajectories and forecast origins.

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This work extends KAHM forecasting from geometric representations and prediction limits to an implemented learning and certification procedure. Kumar and Kargaran (2026) study norm-bounded linear prediction of successor soft coordinates, distinguish exact Koopman closure from empirical fitting, and provide KAHM constructions, fitted-risk upper bounds, and independent-trajectory evaluation. We retain their reconstruction mechanism and independent sampling principle. Our extensions are a common observed-output target, a directly estimable lower bound for Lipschitz decoders, a refinement rule that preserves earlier forecasts, and calibrated decisions about accuracy and fitted improvements. Here a Lipschitz constraint limits how much a decoder's output can change relative to the distance between its inputs. Exact Koopman closure, by contrast, would require one matrix to propagate the feature coordinates exactly throughout the domain; we make no such assumption.

The lower bound has a geometric interpretation. Suppose two future targets are farther apart than the decoder's sensitivity allows their predictions to be. The two prediction errors then cannot both be small. Averaging the squared excess separation over conditionally independent pairs gives a population risk lower bound, which we call the *pairwise obstruction*. The conditioning rule matters: pairing trajectories at a common forecast origin and pairing histories within a reference class generally produce different bounds. Comparisons between bounds must use the same rule.

Refinement also needs a consistent definition of model capacity. Replacing a dictionary can remove earlier predictors. We instead append feature blocks, bound the sum of their coefficient norms, and measure input separation by the largest distance across blocks. Setting new coefficients to zero preserves any old forecast. Duplicate blocks can be merged without increasing the norm budget, so duplication alone changes neither the feasible prediction class nor the pairwise obstruction. A different block can improve the geometry; its benefit to a fitted forecast is evaluated separately on independent data.

The paper makes four contributions.

1. **A KAHM forecasting procedure.** Stable soft associations, retained dictionaries, and constrained multihorizon decoders predict the observed signal. Candidates are fixed before calibration, and the procedure reports an unresolved decision when it cannot certify the requested accuracy.

2. **Matched lower and upper risk bounds.** A pairwise obstruction applies to arbitrary Lipschitz decoders. We compare it with a conditional variance bound, show exactness for a distribution supported on two points, and combine it with fitted-risk upper bounds from independent episodes.

3. **Comparable refinement.** We prove nesting and duplication invariance of the feasible classes, and a necessary amount of distance expansion for a refinement to meet an error tolerance. Paired losses test whether particular fitted refinements improve risk.

4. **Empirical assessment.** Experiments vary soft versus hard associations, direct prediction versus feature rollout, dictionary size, coefficient budget, and input preprocessing. A matched comparison evaluates KAHM, radial basis function (RBF), and affine-hull geometry with three decoder types on all seven tasks, using common preprocessing and dictionary-size options. The analysis separates accuracy, representation size, usefulness of certificates, and variation across fitting repetitions, reporting both gains and regressions.

KAHMs supply the main representation, but the risk and refinement results apply to other fixed representations under the stated assumptions. The guarantees concern expected squared error for a

new episode and a specified origin law. They do not establish that the features form a sufficient state, give conditional prediction-interval coverage, or apply automatically to overlapping windows from a single dependent trajectory.

2 Related work

KAHMs and soft geometric representations. KAHMs construct geometric reconstruction models in reproducing kernel Hilbert spaces (RKHSs) (Kumar et al., 2024, 2025). We use their reconstruction affinities as soft dictionary coordinates for forecasting the observed signal. The reconstruction algorithm is inherited; our development concerns constrained output decoders, refinement with a comparable prediction budget, and decisions based on risk bounds. Soft state aggregation and its adaptive construction predate modern representation learning (Singh et al., 1994). Approximate state abstractions provide guarantees for decision quality in reinforcement learning (Abel et al., 2016), and latent transition models can be learned through prediction objectives relevant to the task (Gelada et al., 2019). In our comparisons, the observed-output target and its evaluation law remain fixed as the representation changes.

Koopman regression and learned dictionaries. Extended dynamic mode decomposition approximates the Koopman operator on a finite dictionary of functions (Williams et al., 2015). Jointly learning the dictionary and its propagation model is also established (Li et al., 2017). Kostic et al. (2022) formulate RKHS Koopman estimation as regression and give finite-sample bounds for independent and mixing data. A coordinate-space residual depends on the chosen basis; Haseli and Cortés (2023) develop a basis-invariant consistency index with a prediction-error interpretation. More recently, Conradie et al. (2026) combine invariance diagnostics, multistep projection-error bounds, and dictionary refinement, while Coote and Colbrook (2026) learn dictionaries using spectral residuals and conditioning penalties. These studies establish forecasting guarantees and refinement methods for learned dictionaries. Our focus is a finite-sample statistical decision about a fixed observed-output target: certify a fitted forecast, exclude an entire class of decoders with a prescribed Lipschitz budget, and preserve admissible forecasts when refining the representation.

Lower bounds and model invalidation. Kumar and Kargaran (2026) give lower confidence bounds for norm-bounded linear prediction of soft successor coordinates. Their bounds compare within-class successor variation with retained input variation. We extend the target and predictor class and study a pairwise lower-bound statistic, called a *witness*, that avoids separately estimating and subtracting two root-variance quantities. The prior paper also supplies upper risk bounds for fitted models and evaluates them on independent trajectories. We inherit that sampling principle. The extensions here are a pairwise witness for the observed forecasting target, refinement geometry, and an implemented selection procedure.

More broadly, Müller et al. (2025) study bounds on the best risk in a model class, and Wu and Yang (2026) give non-asymptotic bounds in terms of minimal empirical risk. Our contribution concerns the pairwise statistic and its implications for a specified representation and sensitivity budget. Pairwise differences also underlie the Gamma test for noise variance in smooth regression (Evans and Jones, 2002). Lipschitz regularity has long been used in set-membership identification (Milanese and Novara, 2004) and in trajectory invalidation through data-driven abstractions (Jin et al., 2020). Blum and Nguyen (2026) use pairwise Lipschitz violations to compute prediction ranges robust to a bounded number of corrupted observations. Those guarantees concern feasible functions

or interpolation under adversarial corruption. Our target is a lower confidence bound on population mean squared forecasting error.

Risk calibration and temporal dependence. Our concentration arguments use classical bounded-mean inequalities (Hoeffding, 1963) and empirical Bernstein bounds (Maurer and Pontil, 2009). Learn then Test formulates predictive risk control as multiple testing (Angelopoulos et al., 2025). We use this principle to select among models fixed before calibration and apply a union bound over the finite candidate family. Non-asymptotic forecasting guarantees also exist for standard statistical models (McDonald et al., 2017).

Our principal experimental protocol uses independent episodes, while permitting dependent observations within an episode. A calibration split from a single dependent series does not itself restore independence. Concentration and decoupling arguments then require additional assumptions, as illustrated by Oliveira et al. (2024) in the distinct setting of conformal coverage. Our confidence statements concern expected squared error. They do not provide prediction intervals or conditional coverage for every observed history.

Scope of the extension. Soft gating and residual fitting have long histories (Jordan and Jacobs, 1994; Bühlmann and Yu, 2003). The block norm used here is also an instance of the established group-norm and multiple-kernel-learning framework (Bach, 2008). Its role is to maintain a comparable coefficient budget as geometric blocks are added, including exact duplicates.

3 Learning and certifying KAHM forecasts

The procedure has three stages. KAHM reconstruction models map each history to geometric features; a constrained decoder maps those features to future observations; and independent calibration assesses the resulting forecasts. Keeping these stages separate also allows alternative representations to use the same decoder and evaluation procedure.

3.1 Notation and basic conventions

Vectors are columns unless they appear as rows of a data matrix; $v^\top$ denotes transpose. For a vector v, $\|v\|_2$ is its Euclidean norm and $\|v\|_1 = \sum_\ell |v_\ell|$. For a matrix M, $\|M\|_{\mathrm{F}} = (\sum_{k,\ell} M_{k\ell}^2)^{1/2}$ is its Frobenius norm and $\|M\|_{\mathrm{op}} = \sup_{\|v\|_2=1} \|Mv\|_2$ is its operator norm. We use $\langle u, v\rangle = u^\top v$ and $\langle M, N\rangle_{\mathrm{F}} = \mathrm{tr}(M^\top N)$ for vector and matrix inner products; tr sums diagonal entries. The symbols $\mathbf{1}_N$, I_N, and e_ℓ denote the all-ones vector, identity matrix, and ℓth coordinate vector, respectively, with the ambient dimension specified by context.

For a real number z, $z_+ = \max\{z, 0\}$ is its positive part. Logarithms are natural; $\lfloor z \rfloor$ is the largest integer not exceeding z. We write $\mathbb{E}$ and $\Pr$ for expectation and probability under the specified sampling law. For a square-integrable random scalar or vector V, $\|V\|_{L^2} = (\mathbb{E}\|V\|_2^2)^{1/2}$; for scalars, the norm inside the expectation is absolute value. An overbar denotes a sample average, and a hat denotes a fitted or estimated quantity. The projection $\Pi_{\mathcal{Y}}(v)$ is the closest point to v in a nonempty closed convex set $\mathcal{Y}$ in Euclidean distance. For a box, this operation clips each coordinate to its allowed interval. All random variables and prediction maps are measurable.

Table 1 collects the main quantities. The forecasting construction and the statistical statements below give their precise domains and sampling assumptions.

Symbol	Meaning
y_t, d_y	Observation at time t and its dimension.
x_t, m	Causal history and number of observed time steps in it.
$\mathcal{H}$, H, q	Set of forecast horizons, its size, and target dimension; for a joint forecast, $q = d_y H$.
Φ_b, C_b	Feature block b and its number of coordinates.
Γ_s, p_s	Stack of the first s feature blocks and total dimension $p_s = \sum_{b=1}^{s} C_b$.
κ, λ	Decoder sensitivity budget and ridge regularization parameter.
$R(f)$, R^*	Population mean squared error of a fitted forecast and infimum over a stated decoder class.
$W_{\kappa,A}$	Population pairwise obstruction, using budget κ and conditioning variable A; distinct from coefficient matrices W_b.
n, K	Number of calibration episodes and number $K = \lfloor n/2 \rfloor$ of disjoint episode pairs.
$Z_{i,jh}$, $Q_{r,jh}$	Loss of candidate j at horizon h on episode i, and obstruction score on pair r.
L_{jh}, U_{jh}	Lower obstruction endpoint and upper fitted-risk endpoint; the unindexed $U = \Gamma(X)$ instead denotes a random representation.
D_h, B_h	Diameter of the known horizon-h output set and squared-loss bound $B_h = D_h^2$.
ϵ_h, δ, α	Requested RMSE tolerance, total failure probability, and failure probability allocated to one confidence statement.

Table 1: Recurring notation. Mean squared error (MSE) is an expected squared Euclidean error; root-mean-square error (RMSE) is its square root. Experiment-specific coordinate averaging is stated separately.

3.2 A fixed raw-output forecasting task

Let $y_t \in \mathbb{R}^{d_y}$ be the observation at integer time t, where $d_y \geq 1$ is the number of observed coordinates. For a history length $m \geq 1$, define the column vector

$$x_t = (y_{t-m+1}^\top, \ldots, y_t^\top)^\top \in \mathbb{R}^{m d_y}.$$

Covariates available at time t may be appended to this vector. A forecast origin is a time t at which a prediction is made using only this available information. Let $\mathcal{H}$ be a finite, nonempty set of positive integer horizons, ordered increasingly, and let $H = |\mathcal{H}|$ be its size. The joint target is

$$Y_t = (y_{t+h}^\top : h \in \mathcal{H})^\top \in \mathbb{R}^q, \qquad q = d_y H.$$

Thus each training row contains all requested future observations in the same order. The theory also considers one horizon at a time, in which case its target dimension is d_y (or one for a separately scored coordinate).

The history length, any fixed target transformation, horizons, and origin sampling rule define the prediction task. Here "raw output" means the observed signal under that fixed transformation, rather than learned feature coordinates. When comparing history lengths, we use the same origins, starting

after the longest history is available. Every candidate is therefore evaluated under the same target and origin law.

The main learner directly predicts each requested horizon, sharing a geometric representation and a joint coefficient budget. It does not generate an entire trajectory by iterating one-step forecasts. The rollout baselines instead iterate a fitted feature-transition model and decode the resulting features. Both approaches are evaluated on the same observed targets; a small one-step feature error alone is not assumed to guarantee an accurate long rollout.

An *episode* is one complete trajectory or other prespecified independent sampling unit. Development data comprise training data, used to fit features and coefficients, and validation data, used to choose hyperparameters or a representative within each model family. Calibration data provide the confidence bounds and may select among candidates frozen before calibration. A separate test set describes forecasting performance after these decisions. Its empirical error estimates population risk; it is not an exact risk value.

3.3 KAHM dictionaries and stable soft associations

A dictionary block describes a history through its associations with a set of reference groups. KAHM associations depend on local reconstruction models. Radial basis function (RBF) associations instead depend on distances to group centers; affine-hull associations use residuals from fitted local affine subspaces together with a locality term. Each construction specifies which histories receive similar coordinates. A soft association retains a weight for every group, while a hard assignment retains only the selected group. These geometric choices need not preserve all information relevant to forecasting.

For block b, partition the training histories into $C_b \geq 1$ reference groups, indexed by $c = 1, \ldots, C_b$, and fit their KAHM reconstruction models as in Kumar et al. (2024, 2025); Kumar and Kargaran (2026). Let $\mathcal{X}$ be the history domain. The resulting score $T_{bc} : \mathcal{X} \to [0, 1]$ measures reconstruction discrepancy, with smaller values indicating a closer match. It is a geometric score, not a class probability.

The folding score combines reconstruction error and angular discrepancy (Kumar and Kargaran, 2026, Section 3.3). Let $z(x) \in \mathbb{R}^{d_x}$ be the preprocessed history, where d_x is its dimension, and let $\widehat{z}_{bcj}(x)$, $j = 1, \ldots, J_{bc}$, be the reconstructions supplied by the $J_{bc} \geq 1$ inherited local models for group c in block b. Write

$$e_{bcj}(x) = 1 - \exp\{-\|z(x) - \widehat{z}_{bcj}(x)\|_2\}, \qquad \theta_{bcj}(x) = \frac{\arccos \gamma_{bcj}(x)}{\pi},$$

where $\gamma_{bcj}(x)$ is the cosine similarity $\langle z(x), \widehat{z}_{bcj}(x)\rangle / (\|z(x)\|_2 \|\widehat{z}_{bcj}(x)\|_2)$, clipped to $[-1, 1]$; when either vector has zero norm, the implementation sets this similarity to zero. The inverse cosine $\arccos$ takes values in $[0, \pi]$. The group score is

$$T_{bc}(x) = \min_{1 \leq j \leq J_{bc}} \left\{ \frac{e_{bcj}(x)^2 + \theta_{bcj}(x)^2}{2} \right\}^{1/2}.$$

Thus both discrepancy terms and the resulting score lie in $[0, 1]$. The inherited reconstruction training is unchanged; this formula specifies how its outputs become the scores used by the forecasting representation.

Fix an exponent $\omega > 0$ and offset $\tau > 0$. The exponent controls the contrast between affinities, and the offset keeps them positive. The reported runs use $\tau = 10^{-6}$. Define

$$\Phi_{bc}(x) = \frac{(1 - T_{bc}(x) + \tau)^{\omega}}{\sum_{r=1}^{C_b}(1 - T_{br}(x) + \tau)^{\omega}}, \qquad \Phi_b(x) = (\Phi_{b1}(x), \ldots, \Phi_{bC_b}(x))^{\top}. \tag{1}$$

Then $\Phi_b(x)$ lies in the probability simplex

$$\Delta_{C_b} = \{v \in \mathbb{R}^{C_b} : v_c \geq 0 \text{ for all } c,\ \mathbf{1}_{C_b}^{\top} v = 1\}.$$

The coordinates express relative reconstruction affinity and need not be calibrated probabilities. To evaluate (1) stably, set $\ell_{bc}(x) = \omega \log(1 - T_{bc}(x) + \tau)$ and $\ell_b^{\max}(x) = \max_c \ell_{bc}(x)$, then compute

$$\Phi_{bc}(x) = \frac{\exp(\ell_{bc}(x) - \ell_b^{\max}(x))}{\sum_{r=1}^{C_b} \exp(\ell_{br}(x) - \ell_b^{\max}(x))}.$$

Subtracting the maximum leaves the normalized weights unchanged. Replacing a small denominator by uniform weights would change the representation, even when relative weights remain numerically resolvable. A retained unit test illustrates this issue: the direct powered affinities sum to less than 10^{-12}, yet one normalized association is essentially one.

The experiment manifest records source hashes for the inherited training and folding routines. The forecasting contribution is the retained representation, constrained output decoder, and calibration procedure. In the original experiments, histories are standardized using training means and variances. Section 6 separately compares training-based range scaling and bounded raw histories. The input subsample and the maximum number of local training points are fixed by each configuration. Once fitted, the feature maps remain fixed during decoder fitting and calibration.

At stage s, retain the first s blocks:

$$\Gamma_s(x) = (\Phi_1(x)^{\top}, \ldots, \Phi_s(x)^{\top})^{\top} \in \prod_{b=1}^{s} \Delta_{C_b} \subset \mathbb{R}^{p_s}, \qquad p_s = \sum_{b=1}^{s} C_b.$$

A *prefix* means these first s blocks. Their cluster partitions need not be nested: preserving earlier forecasts requires retaining the earlier coordinate functions, not their partitions. The experiments append blocks of sizes $2, 4, 8, 16$, giving prefix dimensions $2, 6, 14, 30$.

3.4 A constrained multihorizon decoder

Let N be the number of training histories. Form the response matrix $\mathsf{Y} \in \mathbb{R}^{N \times q}$ with joint targets as rows, and the feature matrix $F = [F_1, \ldots, F_s] \in \mathbb{R}^{N \times p_s}$, where row i of $F_b \in \mathbb{R}^{N \times C_b}$ is $\Phi_b(x_i)^{\top}$. Fit an intercept $a \in \mathbb{R}^q$ and coefficient matrix $W \in \mathbb{R}^{p_s \times q}$ by solving

$$\min_{a,W} \ \frac{1}{N}\|\mathsf{Y} - \mathbf{1}_N a^{\top} - FW\|_{\mathrm{F}}^2 + \lambda \|W\|_{\mathrm{F}}^2 \quad \text{subject to} \quad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} \leq \kappa. \tag{2}$$

Here $W_b \in \mathbb{R}^{C_b \times q}$ contains the coefficient rows for block b, $\lambda \geq 0$ is the ridge parameter, and $\kappa \geq 0$ is the coefficient budget. The intercept is unpenalized. The loss sums squared errors over all target coordinates and horizons, then divides by N; it is not additionally divided by q or H.

For a known output box $\mathcal{Y} \subset \mathbb{R}^q$ containing the target, the reported prediction is

$$f_s(x) = \Pi_{\mathcal{Y}}\big(a + W^\top \Gamma_s(x)\big).$$

The objective in (2) fits the unprojected affine predictor; clipping is applied at prediction time and included in every reported loss. Section 4 writes the same affine map as $a + \sum_b W_b \Phi_b(x)$, using the transposed block convention $W_b \in \mathbb{R}^{q \times C_b}$ to simplify its proofs.

Centering the training matrices eliminates the intercept. Projected accelerated gradient then solves the constrained quadratic. Each projection computes the block Frobenius norms, projects that nonnegative vector onto an ℓ_1 ball of radius κ, and rescales the blocks. The recorded first-order optimality gap bounds the remaining training-objective error; a small change between iterates alone would not give that bound. Appendix A derives both the projection and the gap. The fitted-risk certificate remains valid after early stopping, provided the returned model satisfies the budget used for certification.

The ridge term regularizes fitting, whereas the block constraint defines the feasible sensitivity class. Exact block duplication leaves that class unchanged, but distributing coefficients across duplicates can reduce the squared Frobenius ridge penalty. The replication results therefore concern feasible prediction classes and their population obstruction. The separate numerical test of duplicate-block optimum risks sets $\lambda = 0$ and checks agreement within the reported optimization gaps.

3.5 Calibration and model selection

The candidate family includes history lengths, dictionary prefixes, geometric feature types, and baseline models. Each certificate compares a representation lower bound with a fitted-model upper bound for the same observed target, origin law, and decoder sensitivity budget. Algorithm 1 summarizes the procedure; Section 4 gives the score formulas and their guarantees.

A lower endpoint above tolerance can motivate changing the representation or budget, but does not establish that a particular change will help. Paired episode losses therefore compare old and new fitted forecasts directly. They can certify an improvement even when the representation obstruction is zero. Conversely, the lower bound can exclude a restricted class even if no tested refinement attains the desired accuracy.

In the experiments, cost is the number of retained feature coordinates, with ties resolved by history length and then validation error. This measures representation size. A deployment with unequal feature-evaluation costs can use a different cost function, provided it is fixed before calibration.

4 Pairwise obstructions and certified forecasting

This section establishes lower bounds on the risk achievable from a fixed representation, upper bounds for fitted forecasts, and conditions for representation refinement. All results refer to the same distribution of histories and raw forecasting targets. They require neither a Markov assumption, an invariant distribution, nor exact Koopman closure.

4.1 A directly estimable pairwise obstruction

Let $X \in \mathcal{X}$ be a random available history, let $Y \in \mathbb{R}^q$ be its future target, and let $U = \Gamma(X) \in \mathcal{U}$ be a fixed, measurable representation. Here $\mathcal{X}$ and $\mathcal{U}$ denote the history and representation spaces. A measurable pseudometric $d : \mathcal{U} \times \mathcal{U} \to [0, \infty)$ is symmetric, vanishes on the diagonal, and satisfies

Algorithm 1: Learning and certifying KAHM forecasts

1. Specify horizons, causal origins, known target bounds, RMSE tolerances $\epsilon_h \geq 0$, total failure probability $\delta \in (0, 1)$, and a candidate-cost rule.
2. Use development data to fit geometric blocks and constrained decoders for their retained prefixes. Fit comparison models on the same observed targets.
3. Freeze all candidates, preprocessing choices, sensitivity budgets, and proposed pairs of forecasts to compare.
4. On $n \geq 4$ independent calibration episodes, compute one mean squared loss per episode and horizon. Form disjoint episode pairs and compute obstruction scores at matching origins.
5. Allocate δ across all risk bounds and improvement comparisons. Compute simultaneous lower endpoints L_{jh} and upper endpoints U_{jh} for candidate j at horizon h.
6. Return the least costly candidate satisfying $U_{jh} \leq \epsilon_h^2$ at every required horizon. Report a class as excluded at horizon h when $L_{jh} > \epsilon_h^2$. Mark a candidate–horizon combination inconclusive when neither acceptance nor exclusion applies. If no candidate passes every horizon, return an unresolved decision.

Figure 1: Forecasting and certification for a finite family fixed before calibration. The same procedure applies to alternative feature constructions. A candidate fitted after inspecting calibration needs fresh evaluation data or a separately justified adaptive guarantee.

the triangle inequality; unlike a metric, it may assign zero distance to distinct points. For a sensitivity budget $\kappa \geq 0$, define

$$\mathcal{G}_\kappa(d) = \{g : \mathcal{U} \to \mathbb{R}^q \text{ measurable} : \|g(u) - g(v)\|_2 \leq \kappa d(u, v) \text{ for all } u, v \in \mathcal{U}\},$$
$$R^*_\kappa(\Gamma, d) = \inf_{g \in \mathcal{G}_\kappa(d)} \mathbb{E}\|Y - g(\Gamma(X))\|_2^2.$$

Thus κ limits how much a forecast can change when its representation changes, and R^*_κ is the best population squared-error risk permitted by that limit. For any predictor $f : \mathcal{X} \to \mathbb{R}^q$, write $R(f) = \mathbb{E}\|Y - f(X)\|_2^2$. The representation, pseudometric, and budget are fixed before the calibration data are observed. The class $\mathcal{G}_\kappa(d)$ may contain more decoders than the fitted family; a lower bound on its optimal risk therefore also applies to that family.

To define the pairwise statistic, first draw a conditioning variable A. Then draw (X, Y) and (X', Y') independently from the same conditional law given A, and set $U = \Gamma(X)$ and $U' = \Gamma(X')$. Each copy must have the original marginal law of (X, Y). Taking A to be constant gives an ordinary independent pair. With $z_+ = \max\{z, 0\}$, define

$$W_{\kappa,A}(\Gamma, d) = \frac{1}{2}\mathbb{E}\left[\left(\|Y - Y'\|_2 - \kappa d(U, U')\right)_+^2\right]. \tag{3}$$

The expectation includes both the draw of A and the two conditional draws. We call $W_{\kappa,A}$ a *pair witness*, or *obstruction*: it measures target separation that a decoder with budget κ cannot reproduce

from the representation separation. The variable A specifies a sampling construction. Arbitrary nearest-neighbor matching need not have this conditional independence property.

Theorem 1 (Pairwise prediction obstruction) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$. Then*

$$R^*_\kappa(\Gamma, d) \geq W_{\kappa,A}(\Gamma, d). \tag{4}$$

For a Euclidean representation space with $d(u, v) = \|u - v\|_2$, assume also $\mathbb{E}\|U\|_2^2 < \infty$. Define

$$\sigma^2_{Y|A} = \mathbb{E}\|Y - \mathbb{E}[Y \mid A]\|_2^2, \qquad \tau^2_{U|A} = \mathbb{E}\|U - \mathbb{E}[U \mid A]\|_2^2,$$

where $\sigma_{Y|A}$ and $\tau_{U|A}$ are the nonnegative square roots. These are scalar, averaged conditional variances, rather than functions of a particular value of A. Then

$$W_{\kappa,A}(\Gamma, \|\cdot\|_2) \geq (\sigma_{Y|A} - \kappa\tau_{U|A})_+^2. \tag{5}$$

Both inequalities use the same conditioning variable and evaluation law.

Proof Fix $g \in \mathcal{G}_\kappa(d)$ with finite risk and define the two residuals $r = Y - g(U)$ and $r' = Y' - g(U')$. Since $Y - Y' = r - r' + g(U) - g(U')$, the triangle inequality gives

$$\|Y - Y'\|_2 \leq \|r - r'\|_2 + \kappa d(U, U'), \qquad (\|Y - Y'\|_2 - \kappa d(U, U'))_+ \leq \|r - r'\|_2.$$

Conditional on A, the residuals are independent and identically distributed. Expanding the squared norm therefore yields

$$\begin{aligned}\mathbb{E}[\|r - r'\|_2^2 \mid A] &= 2\mathbb{E}[\|r\|_2^2 \mid A] - 2\mathbb{E}[r \mid A]^\top \mathbb{E}[r' \mid A] \\ &= 2\left\{\mathbb{E}[\|r\|_2^2 \mid A] - \|\mathbb{E}[r \mid A]\|_2^2\right\}.\end{aligned}$$

Square the preceding pointwise bound, take expectations, and divide by two. This gives $W_{\kappa,A} \leq \mathbb{E}\|r\|_2^2 = R(g \circ \Gamma)$. Taking the infimum over g proves (4); predictors with infinite risk satisfy the bound trivially.

For (5), set $a = \|Y - Y'\|_2$ and $b = \|U - U'\|_2$. Pointwise, $a \leq (a - \kappa b)_+ + \kappa b$. For a scalar random variable V, the notation $\|V\|_{L^2} = (\mathbb{E}V^2)^{1/2}$ denotes its root second moment under the pairing law. The triangle inequality in L^2 gives

$$\|(a - \kappa b)_+\|_{L^2} \geq (\|a\|_{L^2} - \kappa\|b\|_{L^2})_+.$$

The same conditional-independence calculation gives $\mathbb{E}a^2 = 2\sigma^2_{Y|A}$ and $\mathbb{E}b^2 = 2\tau^2_{U|A}$. Substitute these identities, square, and divide by two to obtain (5). ■

The witness directly estimates a squared distance discrepancy. It does not estimate two variances separately and then subtract their square roots. This distinction can matter when little input variation remains in the representation. The certificates below use concentration for bounded means; they establish no new minimax estimation rate. Taking a square root of a risk bound can still produce slow convergence near zero. Moreover, (5) compares certificates under a common pairing law. It does not compare pairing at a common time origin with conditioning on a different grouping, such as a reference-class label $c(X)$.

Proposition 2 (Exactness on a two-atom law) *Suppose (U, Y) takes only two values: (u_0, y_0) with probability $p \in (0, 1)$ and (u_1, y_1) with probability $1 - p$. Use Euclidean representation distance and an unconditioned independent pair, and omit the constant conditioning variable from the notation. Then*

$$R^*_\kappa = W_\kappa = p(1-p)\big(\|y_1 - y_0\|_2 - \kappa\|u_1 - u_0\|_2\big)^2_+. \tag{6}$$

An affine map attains this optimum. If forecasts must lie in a nonempty closed convex set containing y_0, y_1, projecting that map onto the set preserves both the optimum and the Lipschitz budget.

Proof Write $a = \|y_1 - y_0\|_2$, $b = \|u_1 - u_0\|_2$, and $\mu = py_0 + (1-p)y_1$. The unequal ordered pairs have total probability $2p(1-p)$, whereas equal pairs contribute zero. Thus (3) gives the right side of (6) as a lower bound.

To attain the bound, first consider $a = 0$: the constant forecast y_0 has zero risk. If $a > 0$, let $v = (y_1 - y_0)/a$ be the unit target direction and let $s = \min\{a, \kappa b\}$ be the allowed fitted separation. Choose

$$f_0 = \mu - (1-p)sv, \qquad f_1 = \mu + psv.$$

Then $y_0 - f_0 = -(1-p)(a-s)v$ and $y_1 - f_1 = p(a-s)v$, so their weighted squared error is $p(1-p)(a-s)^2$, exactly the proposed bound. If $b = 0$, then $s = 0$ and the constant map μ realizes both fitted values. If $b > 0$, use

$$g(u) = f_0 + \frac{\langle u - u_0, u_1 - u_0\rangle}{b^2}(f_1 - f_0).$$

Its linear part has operator norm $\|f_1 - f_0\|_2/b = s/b \le \kappa$, so it is an admissible affine decoder. Both fitted values lie on the segment joining y_0 and y_1. Projection onto the stated convex set fixes them and is nonexpansive, meaning that it cannot increase distances. ■

4.2 Independent episodes, dependent observations within episodes

An *episode* is one complete observed trajectory, or another complete sampling unit specified by the application. Let $E_1, \ldots, E_n$ be independent and identically distributed episodes, also independent of the data used to construct and freeze the models. For horizon $h \in \mathcal{H}$, let $\mathcal{T}_h$ be a fixed, nonempty set of causal forecast origins, available in every episode and used by every compared candidate. Within an episode, observations may be arbitrarily dependent and their distribution may change with time.

Index the J candidates by $j \in \{1, \ldots, J\}$ and write $H = |\mathcal{H}|$ for the number of horizons. Candidate j has a representation Γ_j, a representation pseudometric d_j, and a fixed forecaster $f_{jh} = g_{jh} \circ \Gamma_j$ at horizon h. Its decoder g_{jh} has a known Lipschitz budget κ_{jh}. Let $X_{i,t}$ denote the history at origin t in episode i, and let $y_{i,t+h}$ denote the corresponding single-horizon observation (distinct from the joint target Y_t in Section 3). Define

$$Z_{i,jh} = \frac{1}{|\mathcal{T}_h|}\sum_{t \in \mathcal{T}_h} \|y_{i,t+h} - f_{jh}(X_{i,t})\|_2^2, \qquad R_{jh} = \mathbb{E}Z_{i,jh}. \tag{7}$$

Thus $Z_{i,jh}$ is one average loss per episode, and R_{jh} is its population mean. Equivalently, R_{jh} is the risk for a new episode and an independently, uniformly sampled origin from $\mathcal{T}_h$. It is not a guarantee

conditional on an individual history. Changing the episode weights or the origin distribution changes the risk being certified.

Assume both targets and predictions lie almost surely in a known nonempty closed convex set $\mathcal{Y}_h$. Let $D_h = \sup_{y,z\in\mathcal{Y}_h} \|y - z\|_2 < \infty$ be its diameter, and put $B_h = D_h^2$. This gives the deterministic loss bound $0 \le Z_{i,jh} \le B_h$. Prediction clipping alone does not establish this bound when the target is unbounded. Nor do training minima and maxima establish a population range.

Form $K = \lfloor n/2 \rfloor$ disjoint episode pairs, leaving one episode unused for pairing when n is odd. For $r \in \{1, \dots, K\}$ define

$$Q_{r,jh} = \frac{1}{2|\mathcal{T}_h|} \sum_{t\in\mathcal{T}_h} \Big(\|y_{2r-1,t+h} - y_{2r,t+h}\|_2 - \kappa_{jh} d_j\big(\Gamma_j(X_{2r-1,t}), \Gamma_j(X_{2r,t})\big) \Big)_+^2. \tag{8}$$

For each j, h, the variables $Z_{i,jh}$ are independent across episodes, and $Q_{r,jh}$ are independent across disjoint pairs. Terms within an individual average need not be independent. Their ranges are $[0, B_h]$ and $[0, B_h/2]$, respectively. Write $W_{jh} = \mathbb{E}Q_{r,jh}$. This is exactly the witness in (3) for the horizon-h task when A is a uniformly sampled common origin: conditional on that origin, the two histories and targets come from independent episodes.

For independent, identically distributed scalar observations $V = (V_1, \dots, V_N)$ in a known interval of length $b \ge 0$, define the sample mean and unbiased sample variance by

$$\bar{V} = \frac{1}{N}\sum_{i=1}^{N} V_i, \qquad s_V^2 = \frac{1}{N-1}\sum_{i=1}^{N}(V_i - \bar{V})^2.$$

For $N \ge 2$ and a failure probability $\alpha \in (0, 1)$, use the empirical Bernstein radius

$$\mathrm{rad}_{\mathrm{EB}}(V; b, \alpha) = \sqrt{\frac{2s_V^2 \log(2/\alpha)}{N}} + \frac{7b\log(2/\alpha)}{3(N-1)}. \tag{9}$$

Here $\log$ is the natural logarithm. The bounded-variable inequality of Maurer and Pontil (2009, Theorem 4), also applied to the reflected observations $-V_i$, gives each of the one-sided bounds $\mathbb{E}V_1 \le \bar{V} + \mathrm{rad}_{\mathrm{EB}}$ and $\mathbb{E}V_1 \ge \bar{V} - \mathrm{rad}_{\mathrm{EB}}$ with failure probability at most α. A joint claim for both directions requires allocation to both events.

Theorem 3 (Simultaneous two-sided episode certificate) *Suppose the J candidates and H horizons are frozen using independent development data, $n \ge 4$, and all fitted decoders satisfy their stated Lipschitz budgets. Fix $\delta \in (0, 1)$ and set $\alpha = \delta/(2JH)$. With $\bar{Q}_{jh} = K^{-1}\sum_{r=1}^{K} Q_{r,jh}$ and $\bar{Z}_{jh} = n^{-1}\sum_{i=1}^{n} Z_{i,jh}$, define*

$$L_{jh} = \max\{0, \bar{Q}_{jh} - \mathrm{rad}_{\mathrm{EB}}(Q_{\cdot,jh}; B_h/2, \alpha)\}, \tag{10}$$

$$U_{jh} = \min\{B_h, \bar{Z}_{jh} + \mathrm{rad}_{\mathrm{EB}}(Z_{\cdot,jh}; B_h, \alpha)\}. \tag{11}$$

The dot denotes the full sample over episodes or pairs, so the radius uses sample size K for $Q_{\cdot,jh}$ and n for $Z_{\cdot,jh}$. With probability at least $1-\delta$, conditional on the development data, all candidate–horizon combinations simultaneously satisfy

$$L_{jh} \le W_{jh} \le R_{jh}^* \le R_{jh} \le U_{jh}. \tag{12}$$

Here R^*_{jh} is the infimum risk over all κ_{jh}-Lipschitz decoders of Γ_j, using d_j, the same horizon, the same uniform origin law, and the same raw target.

Proof Condition on the development data so that the entire candidate family is fixed. For each j, h, apply the upper empirical Bernstein bound to the n episode losses and the lower bound to the K pair scores. Since $0 \le W_{jh}$ and $R_{jh} \le B_h$, truncating the endpoints at zero and B_h preserves validity. A union bound over these $2JH$ one-sided events gives $L_{jh} \le W_{jh}$ and $R_{jh} \le U_{jh}$ jointly, with failure probability at most $2JH\alpha = \delta$.

For any admissible decoder, Theorem 1 applies to the joint law obtained by drawing a common uniform origin and two independent episodes. It bounds W_{jh} by that decoder's risk under the same origin mixture. Taking the infimum over these decoders gives $W_{jh} \le R^*_{jh}$. Finally, the fitted decoder belongs to this class, so $R^*_{jh} \le R_{jh}$. The loss and pair statistics may use the same episodes: independence between the upper and lower statistics is unnecessary for the union bound. ■

The probability $1 - \delta$ concerns repeated draws of the calibration episodes. The theorem bounds expected squared error; it does not bound every future error by $\sqrt{U_{jh}}$ or provide a prediction interval with conditional coverage. Its independent sample sizes are n and K, not the number of overlapping windows. Additional origins may reduce the variance of an episode average, but they do not create additional independent episodes. The representation, normalization, dictionary sizes, budgets, horizons, and all baselines included in the joint guarantee must be fixed before calibration. They may depend on the development data because the theorem conditions on those data.

Corollary 4 (Acceptance, exclusion, and inconclusive decisions) *Let $\epsilon_h \ge 0$ be a prespecified root-mean-square error (RMSE) tolerance at horizon h. On the event in* (12)*, $U_{jh} \le \epsilon_h^2$ certifies that the fitted forecaster attains this tolerance. If $L_{jh} > \epsilon_h^2$, no decoder in the corresponding Lipschitz class attains it. When neither condition holds, the decision is inconclusive. Selecting any member of the frozen family that satisfies $U_{jh} \le \epsilon_h^2$ at every required horizon preserves these guarantees. In particular, one may minimize a predeclared cost over the certified candidates and abstain if there are none.*

The selected candidate has the lowest cost among those certified by the data. Other candidates may meet the tolerance without being certified. Also, (12) implies

$$0 \le \sqrt{R_{jh}} - \sqrt{R^*_{jh}} \le \sqrt{U_{jh}} - \sqrt{L_{jh}}.$$

Thus the endpoint gap bounds the fitted decoder's RMSE suboptimality within the stated class. A large gap does not identify optimization error: the witness and either confidence correction may be loose.

For simple sample-size calculations, one may instead prespecify the Hoeffding radius $b\sqrt{\log(1/\alpha)/(2N)}$. If a fitted risk lies at least $\gamma > 0$ below a squared-error tolerance, then

$$n \ge \frac{2B_h^2 \log(1/\alpha)}{\gamma^2}$$

is sufficient for its Hoeffding upper bound to pass that tolerance with probability at least $1 - \alpha$. If the pair witness exceeds the tolerance by γ, then

$$K \ge \frac{B_h^2 \log(1/\alpha)}{2\gamma^2}$$

is sufficient for the corresponding lower bound to exclude the tolerance with that probability. In each case the stated sample size makes the radius at most $\gamma/2$; a one-sided Hoeffding deviation bound controls the remaining $\gamma/2$. These are sufficient sample sizes, not optimal rates. Exclusion depends on the witness margin, rather than on an unknown oracle-risk margin. Choosing the smaller of separately computed Hoeffding and empirical Bernstein bounds requires confidence allocation to both bounds; otherwise the choice must be fixed in advance.

4.3 Refinement that preserves the meaning of the norm budget

Let $\Phi_b : \mathcal{X} \to \Delta_{C_b}$, $b = 1, \ldots, s$, be fixed dictionary blocks, where $\Delta_{C_b} = \{u \in \mathbb{R}^{C_b} : u_c \geq 0,\ \sum_c u_c = 1\}$ is the probability simplex. Retain the entire stack $\Gamma_s = (\Phi_1, \ldots, \Phi_s)$. Refinement appends a block while keeping every previous coordinate function unchanged. For stacked vectors $u = (u_1, \ldots, u_s)$ and $v = (v_1, \ldots, v_s)$ define the block maximum metric $\rho_s(u, v) = \max_{1 \leq b \leq s} \|u_b - v_b\|_2$. Its induced pseudometric on histories and the coefficient norm are

$$d_s(x, x') = \rho_s(\Gamma_s(x), \Gamma_s(x')) = \max_{1 \leq b \leq s} \|\Phi_b(x) - \Phi_b(x')\|_2, \qquad \|W\|_{2,1} = \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}}. \tag{13}$$

Here $\|W\|_{2,1}$ is explicitly a sum over *blocks*, not over individual rows. In this subsection we use column-vector decoders with $a \in \mathbb{R}^q$ and $W_b \in \mathbb{R}^{q \times C_b}$; these coefficient blocks are the transposes of the training matrices in (2).

Let $\mathcal{Y} \subseteq \mathbb{R}^q$ be a nonempty closed convex output set containing Y almost surely. Write $\Pi_{\mathcal{Y}}(z) = \arg\min_{y \in \mathcal{Y}} \|z - y\|_2$ for Euclidean projection; for a box, it clips each coordinate to its allowed interval. Define

$$\mathcal{F}_{s,\kappa} = \left\{ x \mapsto \Pi_{\mathcal{Y}}\left(a + \sum_{b=1}^{s} W_b \Phi_b(x) \right) : a \in \mathbb{R}^q,\ \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} \leq \kappa \right\}. \tag{14}$$

The intercept is unconstrained. Group norm constraints are established tools (Bach, 2008); here the block norm makes the decoder's sensitivity compatible with retained dictionary blocks.

Theorem 5 (Nested and replication-invariant refinement) *Every decoder in $\mathcal{F}_{s,\kappa}$ is κ-Lipschitz with respect to d_s. Appending a block gives $\mathcal{F}_{s,\kappa} \subseteq \mathcal{F}_{s+1,\kappa}$ and $d_{s+1} \geq d_s$ pointwise. Consequently, the infimum prediction risk in this decoder class cannot increase. Neither can the pair witness, provided the budget, target law, and conditional pairing law remain fixed. Appending an exact copy of an existing block leaves both the decoder class and the pair witness unchanged.*

Proof For the unprojected decoder, the triangle inequality and the bound $\|Av\|_2 \leq \|A\|_{\mathrm{F}} \|v\|_2$ give

$$\left\| \sum_b W_b \big(\Phi_b(x) - \Phi_b(x') \big) \right\|_2 \leq \sum_b \|W_b\|_{\mathrm{F}} \|\Phi_b(x) - \Phi_b(x')\|_2 \leq \kappa d_s(x, x').$$

Projection onto a closed convex set is nonexpansive, so this bound also holds for the projected decoder. Assigning zero coefficients to a new block embeds each old decoder in the refined class, proving inclusion and hence nonincrease of the optimal risk. Appending a block can only increase the maximum in d_s. For each paired observation, increasing that distance can only decrease the positive-part discrepancy in (3); taking expectations proves witness monotonicity.

If block $s+1$ exactly copies block b, merge its coefficients into the old block. The prediction is unchanged and $\|W_b + W_{s+1}\|_{\mathrm{F}} \le \|W_b\|_{\mathrm{F}} + \|W_{s+1}\|_{\mathrm{F}}$, so the merged coefficients remain feasible. This proves the reverse class inclusion for duplication. A duplicate also leaves the maximum distance unchanged, and therefore leaves the witness unchanged. ■

These claims concern population optima over nested feasible classes, not the risks of separately fitted finite-sample estimators. They also do not assert that every new block contains useful predictive information. Since the Euclidean diameter of each probability simplex is at most $\sqrt{2}$, $d_s \le \sqrt{2}$ regardless of the number of retained blocks. Certifying a fitted decoder at its smaller actual norm is possible, but changes the comparator class. Refinement monotonicity is asserted only at a common budget.

The choice of norm matters even when no information is added. Let two equally likely histories satisfy $\Phi(x_0) = e_1 = (1,0)^\top$ and $\Phi(x_1) = e_2 = (0,1)^\top$, with scalar targets -1 and 1. Under a single Frobenius constraint of radius one, the largest possible forecast separation is $\sqrt{2}$, and the best affine mean squared error (MSE) is $(2-\sqrt{2})^2/4$. After duplicating the block, the concatenated coefficient vector $(-1/2, 1/2, -1/2, 1/2)$ has norm one and predicts both targets exactly. Duplication has relaxed the effective sensitivity constraint. The block-sum constraint prevents this effect while allowing genuinely new blocks to change the geometry.

Theorem 6 (A necessary amount of geometric change) *Let d_0, d_1 be measurable pseudometrics induced on a common history space by an old and a new representation. Use the same square-integrable target, conditionally independent pairing law, and sensitivity budget $\kappa > 0$. Define the random distance increase $\Delta = (d_1(X,X') - d_0(X,X'))_+$ and assume $\|\Delta\|_{L^2} < \infty$. Let W_0, W_1 be the corresponding witnesses from* (3)*, using these induced history distances. Then*

$$\sqrt{W_1} \ge \left(\sqrt{W_0} - \frac{\kappa}{\sqrt{2}} \|\Delta\|_{L^2} \right)_+ . \tag{15}$$

In particular, if a κ-Lipschitz decoder of the new representation attains RMSE at most $\epsilon \ge 0$, then

$$\|\Delta\|_{L^2} \ge \frac{\sqrt{2}}{\kappa} (\sqrt{W_0} - \epsilon)_+ . \tag{16}$$

On any event $0 \le L_0 \le W_0$, replacing W_0 by L_0 on the right side of (16) *is valid simultaneously for every new representation satisfying these conditions. The new representation need not have been fixed when L_0 was computed.*

Proof Set $a = \|Y - Y'\|_2$ and $v_i = (a - \kappa d_i(X,X'))_+$ for $i = 0, 1$. The positive-part function is nondecreasing and 1-Lipschitz. Consequently, $v_0 \le v_1 + \kappa\Delta$ pointwise, and the L^2 triangle inequality gives

$$\|v_1\|_{L^2} \ge (\|v_0\|_{L^2} - \kappa\|\Delta\|_{L^2})_+ .$$

Since $\|v_i\|_{L^2} = \sqrt{2W_i}$, dividing by $\sqrt{2}$ proves (15). If the new decoder has RMSE at most ϵ, Theorem 1 gives $W_1 \le \epsilon^2$. Combining the two inequalities and rearranging gives (16). Finally, its right side is nondecreasing in W_0. Replacing W_0 by a nonnegative lower bound therefore weakens the necessary condition, for every admissible new representation on the same event. ■

This theorem rules out a proposed repair when its population distance increase is too small. It is a necessary condition, not a sufficient refinement rule or a guarantee that an estimated distance increase is accurate. The quantity L_0 must lower-bound the old *pair witness*; an arbitrary lower bound on old oracle risk cannot be substituted. If the budget also changes from κ_0 to κ_1, the same proof gives the necessary condition

$$\left\|(\kappa_1 d_1(X, X') - \kappa_0 d_0(X, X'))_+\right\|_{L^2} \geq \sqrt{2}(\sqrt{W_{\kappa_0,0}} - \epsilon)_+,$$

where $W_{\kappa_0,0}$ is the old witness at budget κ_0. This form allows the increased separation to come from a larger budget, a changed representation, or both.

4.4 When an added block can improve prediction

The preceding theorem gives a necessary geometric condition. The next calculation gives a sufficient population condition for one proposed feature block: it must correlate with the current residual, and the old decoder must have unused norm budget. A positive obstruction alone implies neither condition.

Proposition 7 (Residual correlation and budget slack) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$ and consider the unprojected affine predictor*

$$f_0(x) = a + \sum_{b=1}^{s} W_b \Phi_b(x), \qquad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} < \kappa.$$

Let $\psi : \mathcal{X} \to \mathbb{R}^p$ be a new feature block with $\mathbb{E}\|\psi(X)\|_2^2 < \infty$. Define

$$r = Y - f_0(X), \qquad s_0 = \kappa - \sum_b \|W_b\|_{\mathrm{F}} > 0,$$
$$C = \mathbb{E}[r\psi(X)^\top] \in \mathbb{R}^{q\times p}, \qquad G = \mathbb{E}[\psi(X)\psi(X)^\top] \in \mathbb{R}^{p\times p}.$$

Here C is the residual–feature cross-moment, G is the feature second moment matrix, and s_0 is the unused budget. If $C \neq 0$, let $a_0 = \|C\|_{\mathrm{F}}^2$ and $b_0 = \mathrm{tr}(CGC^\top)$, where tr sums the diagonal entries. Then $b_0 > 0$, and the coefficient block

$$V = \eta C \in \mathbb{R}^{q\times p}, \qquad \eta = \min\{a_0/b_0,\ s_0/\sqrt{a_0}\}$$

is feasible when appended to the old blocks. Its risk improvement is

$$R(f_0) - R(f_0 + V\psi) = 2\eta a_0 - \eta^2 b_0 \geq \eta a_0 > 0. \tag{17}$$

If Y and $f_0(X)$ belong almost surely to a nonempty closed convex set $\mathcal{Y}$, projecting the new forecast onto $\mathcal{Y}$ retains at least this improvement.

Proof The residual and the new feature block have finite second moments, so C and G are well-defined. Also, $b_0 = \mathbb{E}\|C\psi(X)\|_2^2$. If $b_0 = 0$, then $C\psi(X) = 0$ almost surely, implying

$$\|C\|_{\mathrm{F}}^2 = \langle C, \mathbb{E}[r\psi(X)^\top]\rangle_{\mathrm{F}} = \mathbb{E}[r^\top C\psi(X)] = 0,$$

contrary to $C \neq 0$. Thus $b_0 > 0$ and $\eta > 0$.

For any coefficient matrix V, expand $\|r - V\psi(X)\|_2^2$ and take expectations to obtain

$$R(f_0 + V\psi) = R(f_0) - 2\langle V, C\rangle_{\mathrm{F}} + \mathrm{tr}(VGV^\top).$$

Substituting $V = \eta C$ gives the equality in (17). Since $\|V\|_{\mathrm{F}} = \eta\sqrt{a_0} \le s_0$, appending this block respects the total budget. Since $\eta \le a_0/b_0$, the improvement is at least $\eta a_0 > 0$. Finally, projection onto a closed convex set containing Y cannot increase the new squared error. The old forecast is already in that set, so its reference risk is unchanged. ■

The condition on the old forecast matters. An improvement between two unprojected affine predictors does not by itself imply an improvement between their separately clipped versions. The proposition also makes no claim when the old decoder exhausts its budget. Its moments are population moments: replacing them with training estimates does not establish the same population improvement. For fitted, clipped models, we therefore compare their episode losses directly.

Fix a horizon h and two forecasts before collecting $n \ge 2$ new independent evaluation episodes. Let Z_i^{old} and Z_i^{new} be their losses on episode i, averaged over the same origins as in (7). Define

$$D_i = Z_i^{\mathrm{old}} - Z_i^{\mathrm{new}} \in [-B_h, B_h], \qquad \bar{D} = \frac{1}{n}\sum_{i=1}^{n} D_i, \qquad R^{\mathrm{old}} = \mathbb{E}Z_i^{\mathrm{old}}, \quad R^{\mathrm{new}} = \mathbb{E}Z_i^{\mathrm{new}}.$$

For $\alpha \in (0,1)$, the lower empirical Bernstein bound gives, with probability at least $1-\alpha$,

$$R^{\mathrm{old}} - R^{\mathrm{new}} \ge \bar{D} - \mathrm{rad}_{\mathrm{EB}}(D; 2B_h, \alpha). \tag{18}$$

The interval length is $2B_h$, because D_i can be negative. A positive right side certifies a risk reduction for these two fitted predictors; a nonpositive value is inconclusive. For multiple fixed comparisons, allocate failure probability over the entire comparison family.

Sequential refinement can also be adaptive. At stage s, a candidate may depend on all previous results, provided its evaluation episodes are fresh and independent conditional on that history. Choose stage failure probabilities $\alpha_s > 0$ with $\sum_s \alpha_s \le \delta$, and accept a change only when its paired lower bound is positive. Conditional validity at each stage and a union bound then make every accepted improvement valid on one event of probability at least $1-\delta$. This argument does not permit repeated tuning on a fixed calibration set.

Finally, a positive obstruction need not be removable by refinement. If Y is independent of X and takes values $-a$ and a with equal probability, where $a > 0$, then the minimum risk over all predictors based on X is a^2. Under unconditioned independent pairing, a constant representation has pair witness a^2; no representation of the same observed history removes this independent future noise. Conversely, a zero witness need not imply a useful predictor. For identical deterministic episodes, pairing at the same origin yields zero target distance even when a representation loses all variation between origins. Obstruction, fitted risk, and inconclusive decisions therefore answer distinct questions and should be reported separately.

5 Experimental evaluation

We assess calibration against analytically known risks, forecasting accuracy across representations, certified improvements from additional blocks, and accuracy decisions based on the risk bounds.

Follow-ups examine input scaling (Section 6) and geometric families with matched preprocessing, decoders, and dictionary-size options (Section 7). The release retains every completed fit, candidate, configuration, random stream, and certificate score, including unfavorable and inconclusive outcomes. Throughout this section, MSE denotes mean squared prediction error and RMSE its square root. A test RMSE estimates population RMSE; a certified upper RMSE endpoint is the square root of an upper risk bound.

5.1 Exact-risk calibration and small retained variation

The controlled experiment has two possible input values, called atoms. The first has probability $1-p$, target -1, and representation $u_0 = ((1+a)/2, (1-a)/2)$; the second has probability p, target 1, and representation $u_1 = ((1-a)/2, (1+a)/2)$. Here $a \in [0,1]$ controls how much the representation separates the atoms: $\|u_1 - u_0\|_2 = \sqrt{2}a$. Proposition 2 gives the exact minimum population MSE $p(1-p)(2-\kappa\sqrt{2}a)_+^2$. We also fit the constrained two-atom predictor using 256 independent training observations and compute its population risk analytically. We compare three confidence bounds: the pairwise lower bound, a centered variance-based lower bound for the same raw target, and the upper bound for the fitted model. Each receives a failure-probability allocation of $0.05/3$. Appendix C gives the centered correction and the analytic fitted-risk calculation.

The parameter grid is

$$p \in \{.2, .5\}, \qquad a \in \{0, .2, .6, 1\}, \qquad \kappa \in \{.5, 1, 2\}.$$

Calibration uses $n \in \{64, 256, 1024, 4096\}$ independent observations and 300 independent repetitions per parameter combination, totaling 28,800 trials. One pairwise lower bound exceeds its exact minimum risk; the centered lower and fitted upper bounds have no violations. Thus 28,799 trials satisfy all three inequalities. This finite-grid Monte Carlo check neither proves general coverage nor provides simultaneous confidence across trials. The single violation is retained.

Figure 2 shows the effect of directly estimating the pairwise discrepancy. With $p = .5$ and $\kappa = 1$, the collapsed map $a = 0$ has exact minimum MSE one. At $n = 4096$, the mean pairwise lower endpoint is 0.9177, compared with 0.5327 for the centered endpoint; the mean fitted upper endpoint is 1.0200. The centered input-variance estimate is exactly zero here, but its additive confidence correction is subsequently square-rooted, producing an $n^{-1/4}$-order input penalty. The pair statistic avoids that operation. For $a = .6$ and the same n, the exact minimum is 0.3315, the mean pairwise and centered lower endpoints are 0.2980 and 0.2230, and the mean fitted upper endpoint is 0.3451. These endpoints are in MSE units. At small sample sizes, both lower bounds frequently equal zero. These controlled comparisons do not establish a general minimax rate or uniform finite-sample dominance.

5.2 Bounded time-series forecasting

The main forecasting experiment uses four generators of independent, finite-length episodes: a noisy harmonic oscillator with random frequency and phase, a quasiperiodic signal with two components, a nonlinear autoregression with bounded innovations, and a logistic map with an episode-specific parameter. Their definitions guarantee observations in $[-1, 1]$. Appendix B specifies their equations, initialization, noise ranges, and burn-in conventions. Independence holds between generated episodes; overlapping histories within an episode need not be independent.

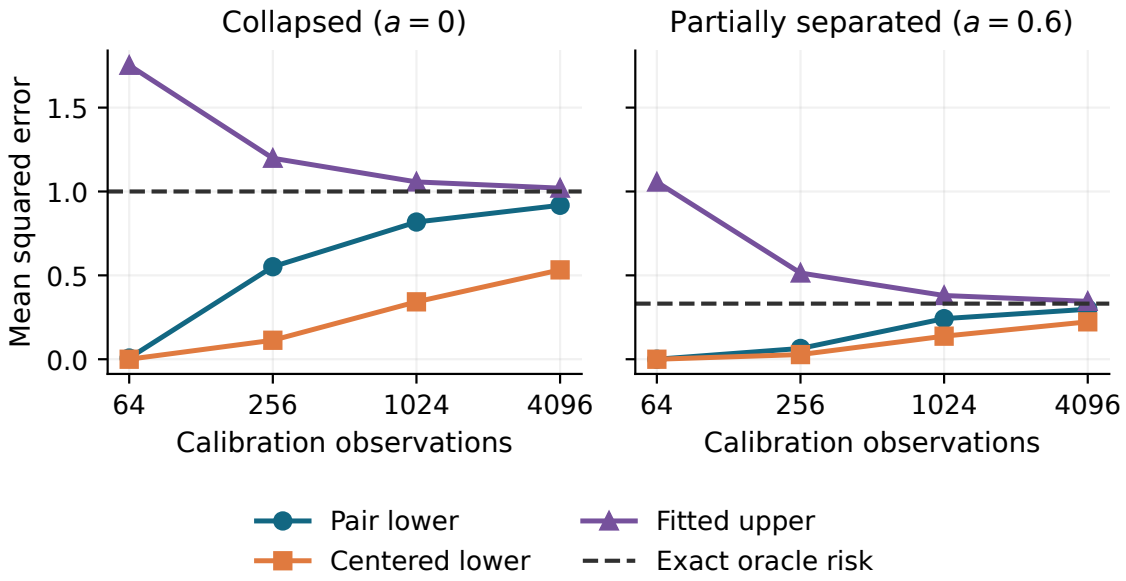


Figure 2: Controlled two-atom experiment with $p = .5$ and $\kappa = 1$. Lines average 300 trials at each calibration size. The minimum risk is known analytically. Both lower bounds use a constant conditioning variable A, so they refer to the same unconditioned pairing law. The upper bound concerns the fitted predictor, whose population risk can exceed the minimum over the class.

For each generator, five complete fitting repetitions use 128 training, 512 validation, 4096 calibration, and 2048 independent test episodes of length 72. Histories have length $m \in \{4, 12\}$, and the forecast horizons are $\mathcal{H} = \{1, 4, 12\}$ time steps. The common zero-based forecast origins are $11, 15, \ldots, 59$, giving 13 origins per episode. All geometric maps are relearned in each repetition. The decoder budget is $\kappa = 4$, the ridge coefficient is $\lambda = 10^{-4}$, and the KAHM affinity exponent is $\omega = 16$. The four successive dictionary blocks have $2, 4, 8, 16$ coordinates; retaining their prefixes gives total feature dimensions $2, 6, 14, 30$. Up to 1600 training histories enter geometry construction and up to 24 local training points enter each KAHM model. The decoder uses all available training rows.

Each repetition freezes 40 candidates before calibration. They include retained KAHM, radial-basis-function (RBF), and local affine-hull features; hard KAHM state assignments; ordinary Frobenius-budget and exact-duplicate ablations; ridge autoregression (AR); direct quadratic regression; affine feature-transition rollouts on KAHM and polynomial dictionaries; persistence; and a constant model. Equal numerical norm budgets do not give different feature maps identical statistical capacities. These fixed configurations assess the procedure; they do not establish the best accuracy attainable by extensive tuning.

The joint failure-probability budget is 0.05 per repetition. We allocate 0.04 to the $2 \times 40 \times 3$ one-sided risk bounds: one lower and one upper bound for every candidate and horizon. This gives $\alpha = 1/6000$ per bound. The remaining 0.01 is allocated over 18 adjacent-prefix improvement comparisons, giving $1/1800$ each. The resulting joint confidence is 95% per complete fitting

Method	Harmonic		Quasiperiodic		Nonlinear AR		Logistic	
	$h = 1$	$h = 12$	$h = 1$	$h = 12$	$h = 1$	$h = 12$	$h = 1$	$h = 12$
KAHM blocks	0.070	0.346	0.078	0.284	0.054	0.074	0.176	0.511
RBF blocks	0.068	0.351	0.068	0.287	0.057	0.074	0.172	0.512
Affine-hull blocks	0.072	0.357	0.110	0.275	0.059	0.074	0.156	0.509
Hard KAHM states	0.097	0.358	0.127	0.300	0.062	0.074	0.235	0.512
Ridge AR	0.038	0.281	0.033	0.259	0.047	0.080	0.409	0.507
Quadratic regression	0.031	0.243	0.030	0.246	0.047	0.081	0.010	0.508
KAHM rollout	0.070	0.358	0.079	0.319	0.055	0.074	0.174	0.524
Polynomial rollout	0.030	0.297	0.030	0.288	0.047	0.084	0.010	0.502
Persistence	0.125	1.085	0.084	0.743	0.055	0.103	0.931	0.661

Table 2: Independent-test RMSE, averaged over five complete fitting seeds. One candidate per family is selected using development validation error averaged over all three horizons, before calibration or test scoring. The full archive includes $h = 4$, seed variability, every candidate, and all confidence endpoints. These means are descriptive estimates, not population-risk certificates.

repetition, rather than across all 20 experimental repetitions. The upper bound uses 4096 episode losses, and the lower bound uses 2048 disjoint episode pairs. The 13 within-episode origins do not multiply these independent sample sizes.

Table 2 reports independent-test errors for one candidate per family, selected by validation MSE averaged over all three horizons. KAHM soft states improve substantially over persistence at long horizons in the harmonic and quasiperiodic tasks, but quadratic and autoregressive models are stronger on these generators. In the nonlinear autoregression, KAHM has mean test RMSE 0.0743 at horizon 12, compared with 0.0802 for ridge AR and 0.0813 for quadratic regression; the simple autoregressive models are better at horizon one. The advantage therefore depends on the task and horizon. Direct KAHM fitting also lowers mean horizon-12 error relative to its iterated feature-transition counterpart on the two periodic tasks. This empirical difference does not establish that direct fitting is always preferable to rollout.

5.3 Useful certificates, class exclusions, and refinement decisions

Table 3 reports the tolerance decisions. Within the retained KAHM family, every nonlinear-AR repetition certifies RMSE at most 0.20 at all three horizons. Every quasiperiodic repetition certifies 0.35, and every harmonic repetition certifies 0.50 but none certifies 0.35. When all comparison models are eligible, the harmonic task also certifies 0.35 in every repetition. No candidate certifies the 0.50 all-horizon requirement for the logistic task. This decision is unresolved: it does not establish that every candidate's true RMSE exceeds 0.50.

Even with thousands of episodes, the selected harmonic quadratic model has mean horizon-one test RMSE 0.0311 but mean upper RMSE endpoint 0.1497. These bounds resolve some tolerances and leave others undecided; coverage alone does not establish practical usefulness.

Of 480 KAHM candidate–horizon statements, 119 have positive lower endpoints, excluding zero risk for their sensitivity-bounded classes on the reported confidence event. Zero endpoints leave adequacy unresolved. Figure 3 compares prefixes using the maximum distance across retained blocks

	KAHM retained family			All candidates		
System	$\epsilon = .20$	.35	.50	$\epsilon = .20$	.35	.50
Harmonic	0/5 (–)	0/5 (–)	5/5 (6)	0/5 (–)	5/5 (12)	5/5 (4)
Quasiperiodic	0/5 (–)	5/5 (14)	5/5 (2)	0/5 (–)	5/5 (4)	5/5 (1)
Nonlinear AR	5/5 (2)	5/5 (2)	5/5 (2)	5/5 (2)	5/5 (2)	5/5 (2)
Logistic	0/5 (–)	0/5 (–)	0/5 (–)	0/5 (–)	0/5 (–)	0/5 (–)

Table 3: Runs certifying the same RMSE tolerance at all horizons $1, 4, 12$. Parentheses give median retained feature dimension among successful least-dimension selections; '–' means no model passed. Cost is a representation-size proxy, not measured runtime. Each run has a joint 95% event covering its complete frozen family and improvement comparisons.

System	$h = 1$	$h = 4$	$h = 12$
Harmonic	16/30	16/30	8/30
Quasiperiodic	5/30	11/30	1/30
Nonlinear AR	0/30	0/30	0/30
Logistic	11/30	4/30	0/30

Table 4: Certified positive risk reductions for adjacent retained KAHM prefixes: three refinements, two history lengths, and five seeds per horizon. The other comparisons are inconclusive; they are not certified degradations.

and budget $\kappa = 4$. Added blocks can change these distances; neither that change nor lower training error guarantees lower population risk.

Paired comparisons certify 72 of 360 adjacent KAHM refinements under the per-repetition joint event (Table 4); the remainder are inconclusive. Nonlinear AR has no certified adjacent-prefix gain despite meeting useful accuracy tolerances. Meeting a tolerance and improving on a previous predictor require different certificates.

Figure 4 isolates replication without training noise. Numerical zero-ridge tests independently check that fitting a single block and exact copies gives matching minimum training risks within the recorded solver gaps. The exact-duplicate candidate in the main study uses zero ridge, whereas the primary models use 10^{-4}. Differences between those fitted candidates are therefore not used as proof of replication invariance.

5.4 Numerical and reproducibility verification

All 20 scheduled main runs completed, retaining all 800 fitted candidates. The largest recorded first-order optimality gap for the block-constrained solver is below 10^{-7}. An independent audit reconstructs all 4800 one-sided risk endpoints and all 360 improvement bounds from the saved double-precision episode scores, with maximum discrepancy zero. A fresh replay of the first harmonic run matches every non-timing result and all 120 retained score arrays exactly, including 983,040 scalar entries. These numerical checks neither replace probability arguments nor formally verify

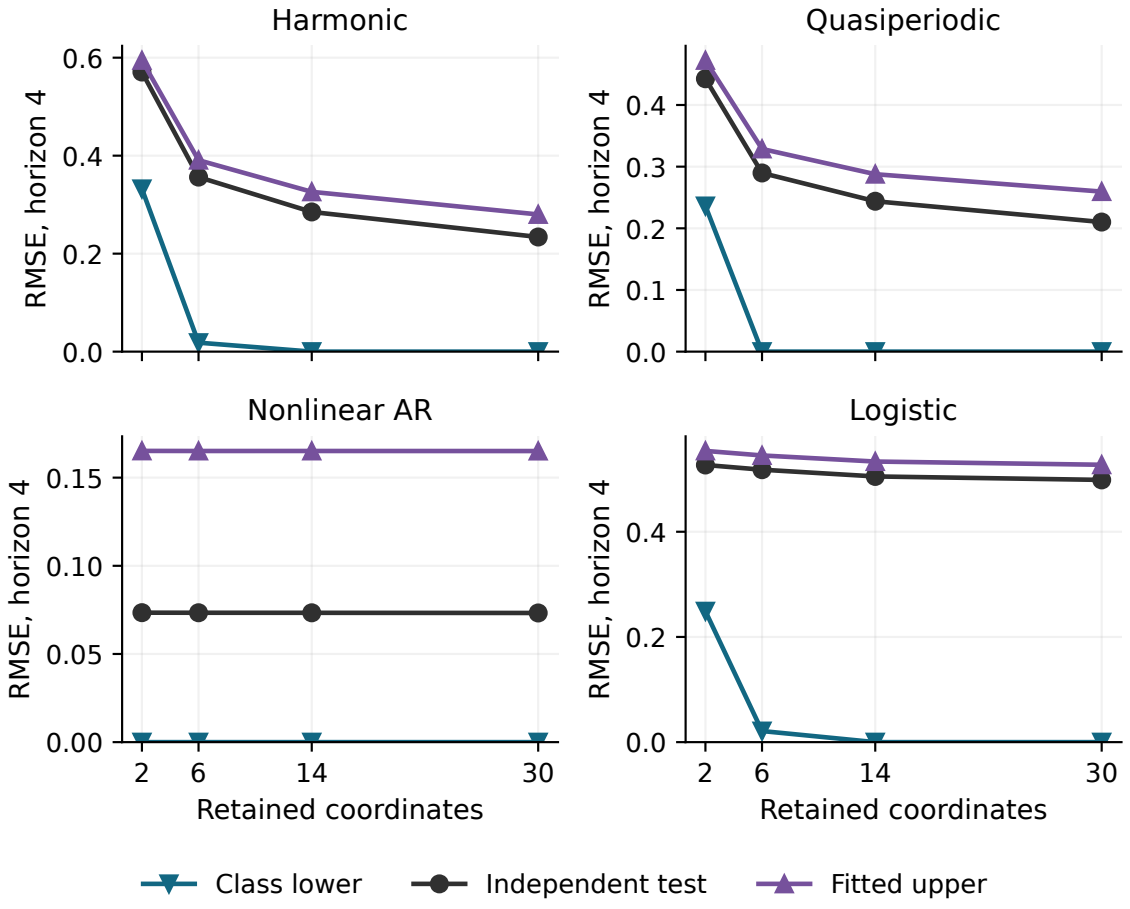


Figure 3: Retained KAHM prefixes with history length 12, sensitivity budget $\kappa = 4$, and horizon four. Curves average five separately learned models. The lower endpoint concerns the whole sensitivity-bounded class; the upper endpoint concerns the fitted model. Test RMSE is an independent finite-sample estimate. Nesting the population prediction classes does not guarantee that every separately fitted model improves.

floating-point arithmetic. Source and configuration hashes, failed-run accounting, and verification output are archived.

5.5 Independently initialized Van der Pol and Duffing episodes

We apply the same frozen procedure to independently initialized Van der Pol and forced Duffing episodes, connecting the experiment to Kumar and Kargaran (2026). Let $x(t)$ denote physical position, $v(t)$ velocity, and t continuous time; a dot denotes differentiation with respect to t. The dynamics are

$$\dot{x} = v, \qquad \dot{v} = (1 - x^2)v - x \qquad \text{(Van der Pol)},$$

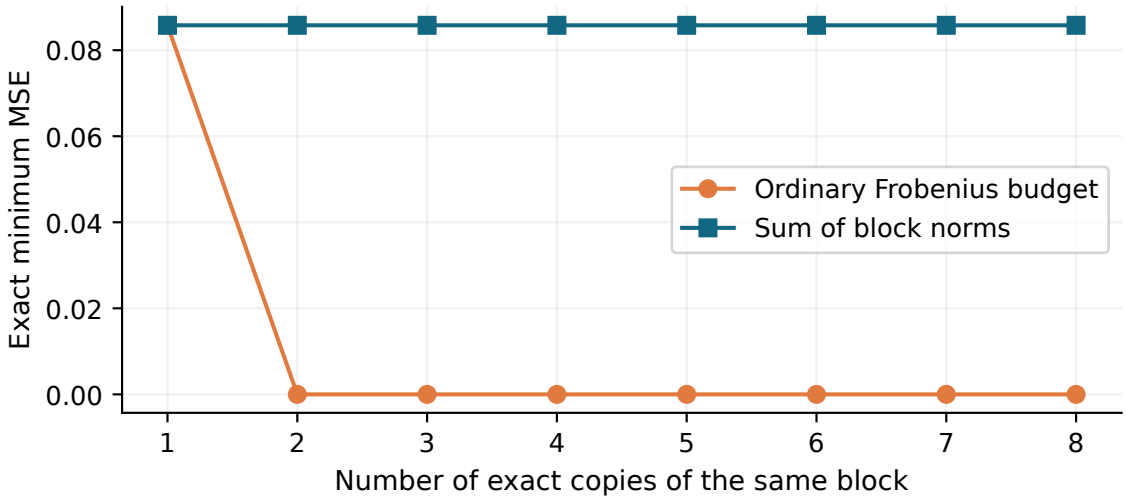


Figure 4: Exact two-atom counterexample at budget one. Copying a two-coordinate feature block permits zero error under an ordinary Frobenius constraint, although the observations contain the same information. The sum-of-block-norms constraint preserves the original minimum risk. The curves are analytic values from Theorem 5, not measured accuracy on a new dataset.

$$\dot{x} = v, \qquad \dot{v} = -0.2v + x - x^3 + 0.3\cos(1.2t + \varphi) \qquad \text{(Duffing)}.$$

Initial position and velocity are independent and uniform on $[-2, 2]$. For Duffing, the forcing phase φ is independently uniform on $[0, 2\pi]$ in each episode. We integrate with classical fourth-order Runge–Kutta at step size 0.025, discard ten time units, and sample 72 observations at interval 0.1. The observed scalar signal is

$$y_j = 0.95\tanh(x(10 + 0.1j)) + \eta_j, \qquad \eta_j \overset{\text{iid}}{\sim} \text{Unif}[-0.02, 0.02], \qquad j = 0, \ldots, 71.$$

Here j is the discrete observation index, $\tanh$ is the hyperbolic tangent, and $\text{Unif}[a, b]$ denotes the uniform distribution on $[a, b]$. The noises η_j are independent and identically distributed (iid). Certificates use the conservative range $[-1, 1]$ for the sensor, whose range lies inside $[-0.97, 0.97]$. They concern sensor-output error, not physical position or velocity, and require neither stationarity nor independent within-episode windows.

Each system has three new complete fitting repetitions, using 128 training, 512 validation, 4096 calibration, and 2048 test episodes. Histories, the 40-candidate family, coefficient budget, and per-repetition joint confidence allocation are unchanged from the main experiment. Horizons $1, 4, 12$ now correspond to $0.1, 0.4, 1.2$ time units. Selection within each family minimizes validation MSE averaged over horizons before calibration or testing. The unchanged forecasting procedure receives the new generator; all candidates, episode scores, and implementation hashes are retained.

The prespecified half-step check compares 16 fixed initializations with identical phases and observation noise at step sizes 0.025 and 0.0125. Maximum sensor differences, 1.69×10^{-6} for Van der Pol and 3.49×10^{-6} for Duffing, are below the fixed 10^{-3} threshold. This assesses numerical consistency, without rigorously bounding global continuous-time integration error.

System	Family	$h = 1$	$h = 4$	$h = 12$	$\overline{\sqrt{U_{12}}}$
Van der Pol	KAHM blocks	0.099 ± 0.001	0.139 ± 0.005	0.381 ± 0.023	0.409
Van der Pol	RBF blocks	0.106 ± 0.005	0.152 ± 0.011	0.383 ± 0.010	0.412
Van der Pol	Affine blocks	0.137 ± 0.003	0.192 ± 0.006	0.299 ± 0.009	0.333
Van der Pol	Ridge AR	0.038 ± 0.000	0.150 ± 0.001	0.353 ± 0.001	0.383
Van der Pol	Quadratic direct	0.032 ± 0.000	0.138 ± 0.001	0.342 ± 0.001	0.374
Van der Pol	KAHM rollout	0.034 ± 0.012	0.064 ± 0.030	0.341 ± 0.033	0.374
Van der Pol	Quadratic rollout	0.029 ± 0.000	0.127 ± 0.002	0.399 ± 0.006	0.427
Van der Pol	Persistence	0.089 ± 0.000	0.339 ± 0.000	0.868 ± 0.001	0.885
Duffing	KAHM blocks	0.080 ± 0.002	0.120 ± 0.004	0.250 ± 0.010	0.300
Duffing	RBF blocks	0.061 ± 0.001	0.098 ± 0.003	0.229 ± 0.007	0.281
Duffing	Affine blocks	0.064 ± 0.004	0.097 ± 0.003	0.223 ± 0.005	0.274
Duffing	Ridge AR	0.019 ± 0.000	0.041 ± 0.000	0.151 ± 0.001	0.214
Duffing	Quadratic direct	0.017 ± 0.000	0.035 ± 0.000	0.137 ± 0.001	0.203
Duffing	KAHM rollout	0.081 ± 0.001	0.126 ± 0.004	0.269 ± 0.014	0.315
Duffing	Quadratic rollout	0.017 ± 0.000	0.035 ± 0.000	0.140 ± 0.002	0.206
Duffing	Persistence	0.033 ± 0.001	0.114 ± 0.002	0.318 ± 0.005	0.363

Table 5: Independent-test sensor RMSE, reported as mean $\pm$ sample standard deviation over three complete fitting repetitions. One candidate per family is selected using validation data. The last column is $3^{-1} \sum_{r=1}^{3} \sqrt{U_{r,12}}$, where $U_{r,12}$ is the horizon-12 upper MSE endpoint for the selected candidate in repetition r. This average summarizes individual certificates; it is not a calibrated guarantee for an ensemble. All individual certificates and all 40 candidates per repetition are archived.

Table 5 shows substantial long-horizon improvement over persistence for retained KAHM models. Mean horizon-12 RMSE is 0.3814 versus 0.8677 on Van der Pol, and 0.2499 versus 0.3185 on Duffing. Other models perform better on these tasks: local affine-hull blocks achieve 0.2990 on Van der Pol, and direct quadratic regression achieves 0.1369 on Duffing. Iterated KAHM has lower mean horizon-12 error than direct KAHM on Van der Pol (0.3408 versus 0.3814), whereas direct KAHM performs better on Duffing (0.2499 versus 0.2686). The earlier advantage of direct fitting on the periodic generators therefore does not extend to every nonlinear oscillator.

The retained KAHM family certifies the all-horizon RMSE requirement 0.35 in all three Duffing repetitions. On Van der Pol, it certifies 0.50 in all three repetitions but does not certify 0.35. When all comparison families are eligible, the 0.35 requirement is certified in every repetition of both systems. No candidate certifies the all-horizon 0.20 requirement in this follow-up. Adjacent KAHM refinements have strictly positive certified improvement in 43 of 54 Van der Pol comparisons and ten of 54 Duffing comparisons, with the same per-repetition confidence allocation.

All six scheduled runs completed, retaining 240 fitted candidates. An independent score audit reconstructs their 1440 one-sided risk endpoints and 108 paired improvement bounds with zero numerical discrepancy. The probability guarantees apply to the specified bounded sensor and independent-episode numerical law. They do not transfer directly to errors in physical state measured under another observation protocol.

System	κ	Features	$h = 1$	$h = 4$	$h = 12$	Passes
Harmonic	1	16.7	0.3737	0.4102	0.4696	0/3
Harmonic	4	30.0	0.1414	0.2313	0.3900	0/3
Harmonic	16	30.0	0.1392	0.2292	0.3875	0/3
Quasiperiodic	1	24.7	0.2510	0.2905	0.3336	0/3
Quasiperiodic	4	30.0	0.1294	0.2050	0.2927	3/3
Quasiperiodic	16	30.0	0.1294	0.2050	0.2927	3/3
Nonlinear AR	1	7.3	0.1059	0.1056	0.1068	3/3
Nonlinear AR	4	30.0	0.0670	0.0730	0.0746	3/3
Nonlinear AR	16	30.0	0.0670	0.0730	0.0746	3/3
Logistic	1	30.0	0.4713	0.5003	0.5052	0/3
Logistic	4	30.0	0.4384	0.4967	0.5044	0/3
Logistic	16	30.0	0.4384	0.4968	0.5044	0/3

Table 6: Sensitivity to the coefficient budget. Test RMSE and retained feature count are means over three new fitting seeds. "Passes" counts validation-selected KAHM predictors certified at tolerance 0.35 simultaneously for $h = 1, 4, 12$. All outcomes are retained. The confidence allocation covers all three budget families within each system–seed repetition.

5.6 Sensitivity to the coefficient budget

The coefficient budget κ defines the prediction class; its numerical value is not directly comparable across feature maps. An unfavorable result at one budget does not establish impossibility at larger budgets. This follow-up was designed after inspecting the main results and uses new development, calibration, and test random streams.

We fix $m = 12$ and compare $\kappa \in \{1, 4, 16\}$ on all four main systems with new seeds 2701–2703. Other settings are unchanged, including 128 training, 512 validation, 4,096 calibration, and 2,048 test episodes per repetition. Each budget has 21 frozen candidates. Within each system and seed, budgets share data streams and geometric blocks; only the coefficient constraint changes. The study comprises 36 runs and 756 models.

For each system, seed, and budget, we select the retained KAHM prefix by mean validation MSE over the three horizons. No budget is selected using test error. We allocate failure probability $0.05/3$ to each budget's complete family of risk bounds and paired comparisons. A union bound gives joint confidence at least 0.95 across the three budgets within each system–seed repetition. This event does not extend jointly across all twelve system–seed repetitions. Table 6 reports test errors and all-horizon certification counts at tolerance 0.35 for the validation-selected predictors.

Increasing the budget from 1 to 4 materially improves fitted KAHM prediction. On the harmonic system, mean one-step RMSE decreases from 0.3737 to 0.1414; on the nonlinear autoregression, it decreases from 0.1059 to 0.0670. On the quasiperiodic system, the number of validation-selected models certified at tolerance 0.35 increases from zero to three out of three. Thus the budget affects accuracy and certification.

Increasing the budget from 4 to 16 produces much smaller changes. Harmonic horizon-12 RMSE changes from 0.3900 to 0.3875, and none of those KAHM models passes the all-horizon 0.35 criterion. Quasiperiodic and nonlinear-AR results are unchanged at the displayed precision. Logistic horizon-12 RMSE remains approximately 0.5044, with no successful all-horizon certification. Thus

increasing the budget alone to 16 leaves limitations in this configuration, without establishing impossibility for richer geometries or larger budgets.

The autoregressive and quadratic baselines fitted at the same numerical budgets are retained in the released results. On the harmonic system at $\kappa = 16$, their mean horizon-12 RMSEs are 0.2377 and 0.2375, respectively, and both certify all three repetitions at tolerance 0.35. On the nonlinear autoregression at $\kappa = 4$, KAHM has smaller horizon-12 RMSE (0.0746) than the autoregressive (0.0800) and quadratic (0.0809) baselines, while its one-step error is larger. These task- and horizon-specific comparisons retain the feature-dependent capacity caveat above.

Every run retains its candidate descriptions, calibration losses, pair scores, and test losses. An independent reconstruction audit passed for all 36 runs, recomputing 2,268 candidate–horizon certificates and 324 paired improvement bounds. All recorded constrained optimizations converged with reported gaps below 10^{-7}. The use of common random streams across budgets was also verified. The checks support the numerical calculations and confidence allocation.

5.7 Writer-disjoint forecasting of original pen trajectories

We evaluate the same forecasters on original PenDigits trajectories (Alpaydin and Alimoglu, 1996). Few writer groups are available, and the risk theorem's independent-episode assumption is unverified.

Data and causal inputs. The University of California, Irvine (UCI) archive provides original time-sampled pen locations and a table of eight spatially resampled locations per digit. We use the original, unnormalized UNIPEN records because the table's whole-character preprocessing is unsuitable for causal forecasting. The acquisition protocol specifies coordinates in $[0, 500]^2$ and a nominal sampling interval of 100 ms. Each recorded coordinate z is transformed to $z/250 - 1$, and scalar predictions are clipped to $[-1, 1]$. We check every source coordinate against the declared acquisition range. There is no per-stroke centering, alignment, normalization, or resampling.

The documented source split contains 7,494 digit records from 30 development writers and 3,498 records from 14 different test writers. We infer writer identity from the second integer in each UNIPEN `COMMENT` field: its range and the approximately 250 records per identifier match the documented writer counts. Because the source does not define these fields explicitly, writer identity remains an inference. Identifiers include their source split. Pen-up gaps are not treated as ordinary time steps.

We retain a digit's first pen-down stroke only if it contains at least 32 samples, and then use its first 32 points. This leaves 1,751 test digits from 13 eligible test writers; one of the original 14 writers has no qualifying first stroke. The population is restricted to writers with eligible strokes. Eligibility depends on completed stroke length, making this a retrospective task that does not cover every newly started stroke. No loss is assigned to the excluded writer. The eligible-writer count was corrected from 14 to 13 before test scoring; the fixed eligibility rule was unchanged. The released records document the correction.

Development and evaluation. Within the development file, the first 24 writer identifiers are used for training and the remaining six for validation. Each of three fixed seeds selects 20 eligible strokes per training writer, for 480 training strokes. All 717 eligible strokes from the validation writers are used for selection. The two coordinates are modeled separately, each from its own scalar history. History lengths are $m \in \{4, 8\}$, and forecast horizons are $\mathcal{H} = \{1, 2, 4\}$ samples. The common

Family	Features	$h = 1$	$h = 2$	$h = 4$
KAHM blocks	30.0	0.0466 ± 0.0010	0.0669 ± 0.0013	0.1127 ± 0.0013
RBF blocks	30.0	0.0421 ± 0.0017	0.0626 ± 0.0019	0.1105 ± 0.0015
Affine-hull blocks	30.0	0.0510 ± 0.0152	0.0664 ± 0.0114	0.1039 ± 0.0053
KAHM, Frobenius budget	30.0	0.0466 ± 0.0010	0.0669 ± 0.0013	0.1127 ± 0.0013
KAHM duplicate blocks	8.0	0.1386 ± 0.0016	0.1532 ± 0.0018	0.1822 ± 0.0018
KAHM hard assignment	16.0	0.0641 ± 0.0004	0.0856 ± 0.0009	0.1297 ± 0.0014
KAHM Koopman rollout	30.0	0.0473 ± 0.0011	0.0688 ± 0.0018	0.1177 ± 0.0025
Ridge autoregression	8.0	0.0130 ± 0.0000	0.0273 ± 0.0000	0.0679 ± 0.0001
Quadratic regression	14.0	0.0077 ± 0.0000	0.0178 ± 0.0001	0.0518 ± 0.0001
Quadratic Koopman rollout	14.0	0.0077 ± 0.0000	0.0185 ± 0.0002	0.0551 ± 0.0005
Persistence	4.0	0.0351 ± 0.0000	0.0696 ± 0.0000	0.1347 ± 0.0000
Constant	1.0	0.2207 ± 0.0007	0.2231 ± 0.0007	0.2273 ± 0.0008

Table 7: Original PenDigits forecasting on 1,751 eligible digits from 13 held-out writers. Entries are coordinate-pooled RMSE, reported as mean $\pm$ sample standard deviation over three fitting seeds. "Features" is the mean retained feature count for the validation-selected models. Each scalar forecast uses only its own coordinate's history. All model selections precede test scoring.

zero-based forecast origins are $7, 11, 15, 19, 23, 27$. All models use causal histories and the same target origins.

For each coordinate and seed, the candidate family contains 40 models: the three soft geometric feature families at four cumulative stages and two history lengths; the KAHM ablations and feature-transition (Koopman) rollouts; autoregressive and quadratic baselines; persistence; and a constant model. Successive geometric block sizes are 2, 4, 8, and 16. Other settings are $\kappa = 4$, ridge coefficient $\lambda = 10^{-4}$, and affinity exponent $\omega = 16$. We select one model per family by validation MSE, weighting writers equally and averaging over the three horizons. All six coordinate–seed families and their selections are frozen before test scoring. Unselected candidates are also retained.

Within each writer, we average squared loss over origins and then eligible digits. The cohort MSE gives each eligible writer equal weight. Let $\widehat{R}_{c,r,h}$ denote this test MSE for coordinate $c \in \{1, 2\}$, fitting repetition $r \in \{1, 2, 3\}$, and horizon h. The coordinate-pooled RMSE in a repetition is

$$\left(\frac{\widehat{R}_{1,r,h} + \widehat{R}_{2,r,h}}{2} \right)^{1/2}.$$

Table 7 reports its mean and sample standard deviation over the three development seeds. These deviations measure fitting variability on the same test cohort, not uncertainty across independent cohorts. A normalized RMSE of 0.01 corresponds to 2.5 acquisition pixels.

Findings and limits. Quadratic regression is the most accurate reported family at all three horizons. Its mean RMSEs are 0.00768, 0.01777, and 0.05184, compared with 0.04661, 0.06686, and 0.11269 for the KAHM block predictor. Ridge autoregression also outperforms the KAHM block predictor. Relative to persistence, KAHM is worse at one step and better at two and four steps. These results do not establish general KAHM superiority. Retained-block and ordinary Frobenius errors are

effectively identical; the selected retained-block norm sums are below $\kappa = 4$, consistent with an inactive constraint.

For completeness, we calculate upper bounds under an additional sampling assumption: eligible writer records are iid from a common writer population and independent of development records. The data documentation does not establish this assumption. Each writer-averaged scalar loss lies in $[0, 4]$. Allocating total failure probability 0.05 over two coordinates, three seeds, 40 candidates, and three horizons gives $2 \cdot 3 \cdot 40 \cdot 3 = 720$ one-sided bounds. With 13 independent writers, the predeclared Hoeffding addition to each empirical MSE is

$$4\sqrt{\frac{\log(720/0.05)}{2 \cdot 13}} = 2.42741.$$

Each resulting upper MSE bound is clipped at 4. Combining the two coordinate bounds by the same averaging-and-square-root rule gives upper pooled-RMSE endpoints of approximately 1.56, far above the observed errors and the reported tolerances of 0.2, 0.35, and 0.5. None of these tolerances is certified. The 1,751 trajectories and their overlapping prediction windows do not supply 1,751 independent writers. This experiment therefore provides descriptive evidence about writer-disjoint forecasting and demonstrates the limited usefulness of generic population certificates with few independent groups.

The release includes the official download URL and archive hash, parser, length-filter accounting, training-record identifiers, frozen model descriptions, and all 9,360 candidate–writer–horizon losses. Recomputing the cohort MSEs from the exported losses agrees to within 1.4×10^{-17}. All recorded block-norm budgets are feasible, and all reported optimization gaps are below 10^{-7}. These checks verify the reported numerical calculation; they do not establish the iid-writer assumption.

6 A follow-up comparison of KAHM input scaling

The preceding experiments standardize history coordinates using their training means and standard deviations. Although raw observations lie in $[-1, 1]$, standardized histories need not. We therefore test whether scaling training histories to $[-1, 1]$ benefits KAHMs, whose reconstruction regularization and folding score depend on input magnitudes and geometry. Target bounds alone do not specify KAHM input scale.

6.1 A fixed intervention and two types of replication

For coordinate j of a history vector x, let x_j denote its value, and let a_j and b_j be the minimum and maximum over all training histories. These extrema are computed before subsampling histories for geometry fitting. The primary alternative to standardization is the min–max transformation

$$z_j(x) = \begin{cases} 2\dfrac{x_j - a_j}{b_j - a_j} - 1, & b_j > a_j, \\ 0, & b_j = a_j. \end{cases} \tag{19}$$

Training extrema remain fixed for validation, calibration, and testing; none are estimated from these data or a completed test trajectory. Unseen values can map outside $[-1, 1]$; this primary preprocessing variant, or *arm*, does not clip them. A diagnostic arm applies $\widetilde{z}_j(x) = \max\{-1, \min\{1, z_j(x)\}\}$, and another uses bounded raw histories. Constant training coordinates map to zero in both min–max arms. Together with standardization, these give four arms.

	Original fitting seeds				New fitting seeds			
Task	Standard	Min–max	$\Delta\%$	Wins	Standard	Min–max	$\Delta\%$	Wins
Harmonic	0.049148	0.048150	-2.03	4/5	0.050240	0.049586	-1.30	3/3
Quasiperiodic	0.037040	0.036182	-2.32	3/5	0.038290	0.035842	-6.39	3/3
Nonlinear AR	0.004561	0.004569	+0.18	4/5	0.004718	0.004656	-1.31	2/3
Logistic	0.164577	0.168931	+2.65	1/5	0.164632	0.176103	+6.97	0/3
Van der Pol	0.058343	0.060572	+3.82	0/3	0.054809	0.056049	+2.26	1/3
Duffing	0.027776	0.025812	-7.07	3/3	0.027913	0.025877	-7.29	3/3
PenDigits	0.006449	0.005173	-19.79	3/3	0.006792	0.005457	-19.65	3/3

Table 8: Validation-selected retained KAHM predictors: mean test MSE over all three horizons and fitting seeds. The primary alternative is training min–max scaling without query clipping. $\Delta\% = 100(\mathrm{MSE}_{\mathrm{minmax}}/\mathrm{MSE}_{\mathrm{standard}} - 1)$; negative values denote improvement. Wins counts fitting repetitions with a smaller all-horizon MSE. Original main tasks have five repetitions; original oscillators and PenDigits have three. All new-seed comparisons have three. Synthetic confirmation uses new development, calibration, and test episodes. PenDigits confirmation uses new training subsamples and the same held-out writer cohort, with coordinate MSE pooled before averaging horizons. Original-seed results reuse previously evaluated streams and are descriptive. No average across different task units is taken.

The intervention changes only the inputs to KAHM geometry and to models built from it: retained-block, single-Frobenius-budget, duplicate-block, hard-assignment, and feature-rollout models. The non-KAHM feature families, forecast targets, output clipping, coefficient budgets, ridge coefficient, affinity exponent, history lengths, origins, sample counts, local-point limits, and random streams remain unchanged. Within each arm, each model family selects its history length and dictionary stage by mean validation MSE across horizons. Test error is not used to choose among preprocessing arms.

The first comparison pairs all four arms on identical data for every original fitting seed and coefficient-budget condition. The scaling hypothesis followed inspection of the original study, so these reused streams are not a new test set. The primary outcome, alternatives, selection rule, and schedule were recorded before examining the scaling results; all outcomes, including regressions, are retained. The exact-risk two-atom experiment and analytic duplicate-invariance illustration do not fit KAHM input geometry and therefore admit no scaling intervention.

For confirmation, we fixed the same four arms and used fitting seeds 4701–4703 on all six synthetic systems. The original stream-offset rule then generates new training, validation, calibration, and test episodes, with the original sample counts and main-study budget $\kappa = 4$. This adds 18 paired synthetic repetitions, without tuning between the replay and confirmation stages. For PenDigits, the same three new seeds select new training subsamples from the existing development writers. The validation writers and held-out writer cohort remain unchanged. Thus the PenDigits confirmation assesses variation across training subsamples on a fixed cohort, rather than variation across newly sampled writers.

	Original fitting seeds		New fitting seeds	
Task	$\overline{\Delta} \pm s_\Delta$	$\mathrm{SE}_{\mathrm{ep}}$	$\overline{\Delta} \pm s_\Delta$	$\mathrm{SE}_{\mathrm{ep}}$
Harmonic	-0.9977 ± 1.2991	0.0360	-0.6544 ± 0.1582	0.0198
Quasiperiodic	-0.8578 ± 1.8211	0.0488	-2.4482 ± 1.3631	0.0630
Nonlinear AR	$+0.0083 \pm 0.0786$	0.0089	-0.0619 ± 0.0935	0.0151
Logistic	$+4.3538 \pm 5.1613$	0.1365	$+11.4711 \pm 6.6749$	0.2367
Van der Pol	$+2.2297 \pm 1.6634$	0.1242	$+1.2396 \pm 4.2734$	0.1101
Duffing	-1.9642 ± 1.4355	0.0950	-2.0360 ± 0.3666	0.0692
PenDigits	-1.2760 ± 0.4998	–	-1.3346 ± 0.1942	–

Table 9: Two distinct sources of variation for the primary scaling comparison. All numbers are in units of 10^{-3} MSE for the corresponding task. Δ is min–max minus standard MSE after averaging horizons within a repetition; s_Δ is its sample SD across complete fitting repetitions. $\mathrm{SE}_{\mathrm{ep}}$ measures paired test-episode Monte Carlo uncertainty of the mean difference, conditional on the fitted, validation-selected models. It does not measure training variability. No iid-writer SE is assigned to the fixed PenDigits cohort. Original-seed quantities reuse previously evaluated streams and are descriptive. Neither column is a finite-sample certificate of an algorithm-level gain.

6.2 Accuracy gains, regressions, and fitting variation

Table 8 reports the primary comparison for all tasks. Mean all-horizon MSE decreases on the harmonic, quasiperiodic, and Duffing systems in both stages. The Duffing reduction is 7.07% on the original seeds and 7.29% on the new seeds. The nonlinear AR effect is small and changes sign between stages. Logistic error increases by 2.65% and 6.97%, and Van der Pol error increases by 3.82% and 2.26%. An improvement averaged over horizons need not hold at every horizon. For example, on the original quasiperiodic repetitions, short-horizon gains accompany an increase in horizon-12 mean RMSE from 0.2841 to 0.2894.

Table 9 distinguishes complete fitting repetitions from test episodes evaluating a fitted model. Let Δ be min–max MSE minus standardized-input MSE per repetition, with mean $\overline{\Delta}$ and sample standard deviation s_Δ across repetitions. The standard error $\mathrm{SE}_{\mathrm{ep}}$ accounts for finitely many independent test episodes, conditional on the fitted, validation-selected models. We average paired loss differences over horizons within each episode, estimate the variance of each repetition's mean difference, and combine these variances for the mean across repetitions. Many episodes can yield a small conditional standard error without providing many independent fits.

The mean MSE change can oppose the majority of seed-specific effects because their magnitudes differ. Every seed's sign and effect size are archived. Neither a small conditional standard error nor a favorable mean establishes improvement for a newly fitted model.

The retained soft KAHM predictor has its largest relative gain on PenDigits: cohort MSE, averaged over coordinates and horizons, decreases by 19.79% in the replay and 19.65% with the new training subsamples. Table 10 reports horizon-specific RMSE and its variation across fitting seeds. All three horizons improve for each coordinate and each fitting seed in both stages. Every held-out writer also has a smaller mean loss, averaged over coordinates and horizons, in each seed. These comparisons repeatedly evaluate the same 13 writers and do not establish a claim about a population

Fitting seeds	Input preprocessing	$h = 1$	$h = 2$	$h = 4$
1701–1703	Standardization	0.04661 ± 0.00098	0.06686 ± 0.00134	0.11269 ± 0.00132
1701–1703	Min–max	0.03693 ± 0.00228	0.05667 ± 0.00257	0.10454 ± 0.00300
4701–4703	Standardization	0.04816 ± 0.00199	0.06884 ± 0.00223	0.11536 ± 0.00202
4701–4703	Min–max	0.03950 ± 0.00123	0.05922 ± 0.00128	0.10631 ± 0.00090

Table 10: PenDigits retained KAHM forecasting: mean $\pm$ sample SD of coordinate-pooled RMSE over three fitting seeds. Coordinate MSE is pooled before taking each seed's square root. All 1,751 eligible digits and 13 held-out writers are identical across both seed sets and preprocessing arms. The SD measures fitting variation on this cohort, not uncertainty across independent test cohorts. A normalized error of 0.01 corresponds to 2.5 acquisition pixels.

of independently sampled writers. Quadratic regression and ridge AR remain more accurate than the scaled retained KAHM predictor on this task; its one-step error also remains above persistence.

The gain does not extend to every predictor built from KAHM features. On PenDigits, the hard-assignment model's all-horizon MSE increases by 3.96% on the original seeds and 4.14% on the new seeds, with degradation in every fitting repetition. The complete records also retain the other KAHM ablations and rollouts. These results support a scaling benefit for particular soft representations under the stated protocol, rather than for all KAHM models.

As a separate secondary outcome, KAHM feature rollout has smaller mean all-horizon MSE on 7 of seven original tasks and 6 of seven new-seed tasks. The new-seed reductions include 25.28% on Van der Pol, 8.47% on Duffing, and 19.83% on PenDigits. Logistic rollout instead worsens by 3.61%.

6.3 Diagnostic arms, coefficient budgets, and certification

Clipped and unclipped min–max scaling give similar retained KAHM results; raw histories can change accuracy differently (Table 11). Both diagnostic arms retain the logistic and Van der Pol limitations, so histories lying in $[-1, 1]$ do not ensure improvement. Raw histories remove centering and variance normalization; min–max scaling changes location and coordinate scale. The study does not isolate individual components of KAHM geometry.

Table 12 reports every original coefficient-budget condition. At $\kappa = 1$, the mean effects are small and the signs differ across seeds. At $\kappa = 4$ and 16, the quasiperiodic comparison improves in all three repetitions, while the logistic comparison worsens in all three. The harmonic mean change has different signs at different budgets. Neither the budget nor the preprocessing arm is selected by test performance. These results limit extrapolation of a gain from one coefficient class to another.

At RMSE tolerances 0.20, 0.35, and 0.50, standardization and primary min–max scaling give equal counts of repetitions whose retained KAHM upper endpoints meet the requirement at every horizon, for every original synthetic system and budget and every synthetic confirmation system. PenDigits retains its small writer cohort and uninformative nominal endpoints. Failure allocations cover multiplicity within each arm, but not selection across arms. A confidence interpretation requires arm specification independent of calibration outcomes. Fixing the replay protocol before rerunning previously used streams does not establish this independence; replay endpoints are nominal. Fresh

	Original fitting seeds		New fitting seeds	
Task	Clipped min–max	Raw histories	Clipped min–max	Raw histories
Harmonic	-2.03	-1.59	-1.30	-1.79
Quasiperiodic	-2.31	-3.96	-6.39	-6.75
Nonlinear AR	+0.18	-0.74	-1.31	-1.52
Logistic	+2.65	+3.87	+6.97	+4.80
Van der Pol	+3.82	+5.27	+2.26	+4.79
Duffing	-7.07	-13.30	-7.30	-11.59
PenDigits	-19.78	-18.51	-19.64	-14.57

Table 11: Diagnostic preprocessing arms, reported as percent change in all-horizon test MSE relative to standardization for the retained KAHM family. Negative values denote improvement. Clipped min–max fixes all query inputs inside $[-1, 1]$; raw histories use the task's existing bounded observations without standardization. These test comparisons are descriptive and are not used to choose a preprocessing arm per task. New-seed PenDigits results use the same writer cohort.

	$\kappa = 1$		$\kappa = 4$		$\kappa = 16$	
Task	$\Delta\%$	Wins	$\Delta\%$	Wins	$\Delta\%$	Wins
Harmonic	+0.41	2/3	+0.17	2/3	-1.24	2/3
Quasiperiodic	-1.16	1/3	-4.94	3/3	-4.94	3/3
Nonlinear AR	+0.01	1/3	-0.59	2/3	-0.59	2/3
Logistic	-0.13	1/3	+1.89	0/3	+1.87	0/3

Table 12: Scaling comparison within the original coefficient-budget sensitivity protocol: history length 12, seeds 2701–2703, and the same data across budgets. Entries are the percent change in mean all-horizon test MSE for training min–max scaling relative to standardization, plus the number of improving fitting seeds. Every stage is selected by mean validation MSE within its arm and budget. Negative values denote improvement. This comparison reuses the original sensitivity streams and is descriptive; the six-system confirmation uses new synthetic episodes.

synthetic confirmation episodes support the stated within-arm guarantees. New PenDigits fitting seeds still reuse previously examined writers. The cross-arm accuracy comparisons therefore do not certify preprocessing improvements.

Reproducibility and interpretation. The standardization replay exactly reproduces all retained non-timing results. Across arms, non-KAHM models retain identical numerical results and saved synthetic score arrays; writer-level control losses are also unchanged. Four truncated score archives—two per stage—were regenerated with the same source, configuration, and seeds; non-timing records and candidate descriptions agree exactly. Damaged and regenerated file hashes and replay records are retained. Reporting followed archive-integrity checks and independent aggregation of the selected models' test scores.

Standardization remains the reference preprocessing for the original experiments. The repeated PenDigits and Duffing gains, together with the logistic and Van der Pol regressions, show that the benefit of input scaling depends on the task and prediction architecture.

7 A matched comparison of KAHM and alternative geometries

The preceding experiments do not fully separate geometry and decoder effects: the KAHM Frobenius-budget and rollout families lack corresponding radial basis function (RBF) and affine-hull controls. We compare all three geometries with identical decoder choices and validation opportunities on all seven tasks. The protocol was specified before fitting, informed by the earlier experiments.

7.1 Design and validation selection

Each geometry is paired with three prediction architectures, referred to as *heads* in the tables. *Block direct* predicts the requested horizons under the sum-of-block-Frobenius-norms constraint in (2). *Frobenius direct* uses a single Frobenius norm constraint on the concatenated coefficient matrix. *Rollout* fits an affine transition between feature vectors, iterates this transition, and decodes the resulting vector into the observed signal (Appendix B). Every geometry–head pair has access to the four retained dimensions $2, 6, 14, 30$, both history lengths, and all four preprocessing options from Section 6. Unlike the preceding scaling study, this comparison varies preprocessing for KAHM, RBF, and affine-hull features alike.

Training data, horizons, origins, regularization, geometry subsampling, and target units follow the original configurations. Direct heads share a numerical coefficient budget, which does not make feature maps statistically equivalent. Rollout sensitivity budgets use the fitted transition and output decoder. Equal feature counts need not imply equal runtime or kernel-evaluation cost.

The five fitting seeds 5701–5705 generate new training, validation, calibration, and test episodes for every synthetic task. PenDigits uses new training-stroke subsamples and geometry-fitting randomness within the same development and held-out writer cohorts, measuring fitting variation without new test writers.

For each task, seed, geometry, and head, the primary analysis selects preprocessing, history length, and retained dimension by mean validation MSE across horizons. Ties are resolved by feature count, history length, preprocessing order (standardization, min–max, clipped min–max, raw), and candidate name. This gives six primary comparisons per task and seed: KAHM against each of the two alternative geometries under each of the three heads. The selected dimension, history, and preprocessing can differ between geometries, but the available choices are identical.

The descriptive outcome is test MSE averaged over horizons; percentage changes use ratios of seed-averaged MSEs, not means of seed-specific percentages. Every paired seed effect is retained; tasks are not pooled. Additional curves fix the dimension and select history length and preprocessing by validation. A secondary analysis also selects the head by validation; it appears in `taskwise_comparisons.csv`, in rows marked `analysis=head_selected` and `horizon=all`; Appendix B.6 gives the analysis directory.

PenDigits selects each coordinate's predictor separately and weights validation writers equally. After selection, coordinate losses and horizon losses are averaged within each held-out writer, and the 13 writers receive equal weight. Repeated fitting seeds do not provide additional independent writers.

7.2 Confidence accounting and interpretation

For each synthetic task and seed, each arm receives failure budget 0.01: the inherited $80/20$ allocation assigns 0.008 to fixed candidate risk bounds and 0.002 to refinement comparisons. A union bound over four arms uses 0.04; the remaining 0.01 is divided equally among the six primary geometry comparisons.

For each primary comparison, we subtract KAHM squared loss from the alternative's squared loss and average the difference over horizons within each independent calibration episode. The resulting episode averages lie in $[-4, 4]$. Applying the paired empirical Bernstein bound (18), with range length 8 and failure probability $0.01/6$, gives a lower confidence bound on the alternative's risk minus KAHM's risk. A positive lower endpoint certifies lower risk for that fitted KAHM predictor. All validation selection occurs before calibration and uses independent development data.

This allocation gives a simultaneous 95% event within each synthetic task–seed combination, not across tasks, seeds, or earlier campaigns. Fixed-dimension and head-selected comparisons remain descriptive; an inconclusive improvement certificate does not establish equal risks.

PenDigits retains nominal writer-level upper endpoints with failure budget 0.0125 per preprocessing arm, allocated jointly over candidates, coordinates, fitting seeds, and horizons. The allocations sum to 0.05. A joint confidence interpretation requires both independent and identically distributed writers and candidate specification independent of the writer losses used for calibration. The first assumption remains unverified; the second is not established for this follow-up because the design was informed by earlier results on the same cohort. Consequently, these endpoints do not provide a new confirmatory population guarantee. No additional independent writer cohort is available for the geometry comparisons, and paired writer differences remain descriptive.

7.3 Geometry, prediction architecture, and task interact

All scheduled fits completed, yielding 13,440 retained candidates. Table 13 reports all 42 primary task–head–alternative contrasts: KAHM has lower mean test MSE in 28 and higher mean test MSE in 14. These contrasts share tasks and fitted models; they are not independent evidence for a universally preferred geometry. Table 14 gives absolute errors for all nine geometry–head families.

On nonlinear AR, KAHM improves on both geometric alternatives under every head in all five fitting seeds. Its block-direct mean MSE is 0.00456, compared with 0.00511 for ridge AR and 0.00523 for quadratic regression. On PenDigits, the KAHM reductions relative to RBF are 15.6%, 17.2%, and 17.9% for block direct, Frobenius direct, and rollout, respectively, with lower loss in all five seeds. The corresponding mean reductions relative to affine-hull features are 12.7%, 1.9%,

	Block direct		Frobenius direct		Rollout	
Task	RBF	Affine	RBF	Affine	RBF	Affine
Harmonic	-2.4 (5)	-5.1 (5)	-1.5 (5)	$+6.1$ (0)	-0.2 (3)	$+1.2$ (1)
Quasiperiodic	-2.0 (2)	-1.4 (2)	-1.7 (3)	$+30.1$ (0)	-6.1 (2)	$+3.3$ (1)
Nonlinear AR	-2.7 (5)	-4.4 (5)	-2.7 (5)	-4.4 (5)	-5.0 (5)	-4.9 (5)
Logistic	-1.0 (3)	-4.5 (5)	-0.4 (2)	-5.8 (5)	$+1.7$ (1)	-5.7 (5)
Van der Pol	-14.4 (5)	$+9.5$ (2)	-14.5 (3)	$+102.7$ (0)	-17.1 (4)	$+92.8$ (0)
Duffing	$+3.8$ (3)	$+13.1$ (1)	$+10.4$ (0)	$+70.6$ (0)	$+1.4$ (3)	$+79.9$ (0)
PenDigits	-15.6 (5)	-12.7 (4)	-17.2 (5)	-1.9 (4)	-17.9 (5)	-1.4 (3)

Table 13: Matched geometry follow-up: percentage change in test MSE of KAHM relative to the indicated alternative, within each head. Negative values favor KAHM. Parentheses give the number of 5 fitting seeds with lower KAHM MSE. Preprocessing, history and retained dimension are selected separately for each geometry and head by validation MSE only. Percentages use ratios of MSE averaged over horizons and fitting seeds within each task; PenDigits first pools coordinate losses and equally weights eligible writers. Synthetic calibration comparisons use new episodes and a separate multiplicity allocation. The PenDigits comparisons reuse the same cohort. All test effects in this table are descriptive.

and 1.4%, with four, four, and three favorable seeds. This advantage is specific to the geometric comparison. PenDigits Frobenius-direct KAHM MSE is 0.00504, while ridge AR and quadratic regression achieve smaller MSEs of 0.00184 and 0.00102.

The oscillator controls limit how much of an accuracy gain can be attributed to KAHM geometry. On Van der Pol, KAHM mean MSE is 14.4–17.1% lower than RBF across the three heads, yet exceeds affine-hull MSE by 9.5%, 102.7%, and 92.8%, respectively. On Duffing, the corresponding excesses over affine-hull MSE are 13.1%, 70.6%, and 79.9%. The affine-hull advantage under Frobenius-direct and rollout heads occurs in every fitting seed on both oscillators. A gain from changing a KAHM decoder or its preprocessing therefore does not by itself establish a benefit of KAHM geometry. Mean effects can also disagree with most seed-specific effects: the favorable quasiperiodic block-direct means against both alternatives occur with lower KAHM loss in only two of five seeds.

Figure 5 reports every fixed-dimension comparison. On PenDigits, KAHM has lower mean MSE than RBF at all four dimensions under each head, so the advantage cannot be explained solely by a larger selected feature count. Its ranking against affine-hull features changes with dimension. On nonlinear AR, all three KAHM heads improve on both alternatives at dimensions 14 and 30, while the smaller dictionaries have mixed effects. Theorem 5 concerns population optima for fixed retained blocks. These curves also reselect history length and preprocessing at each dimension and therefore do not test that theorem.

Only 1 of the 180 planned synthetic calibration comparisons has a positive improvement lower endpoint: Van der Pol, seed 5703, KAHM rollout versus RBF rollout, with a lower bound of 0.00107 on the all-horizon MSE reduction. The other 179 comparisons remain inconclusive. This positive statement concerns those two fitted predictors within the declared simultaneous event for that repetition. It is distinct from the descriptive test-effect table and gives no population comparison for PenDigits.

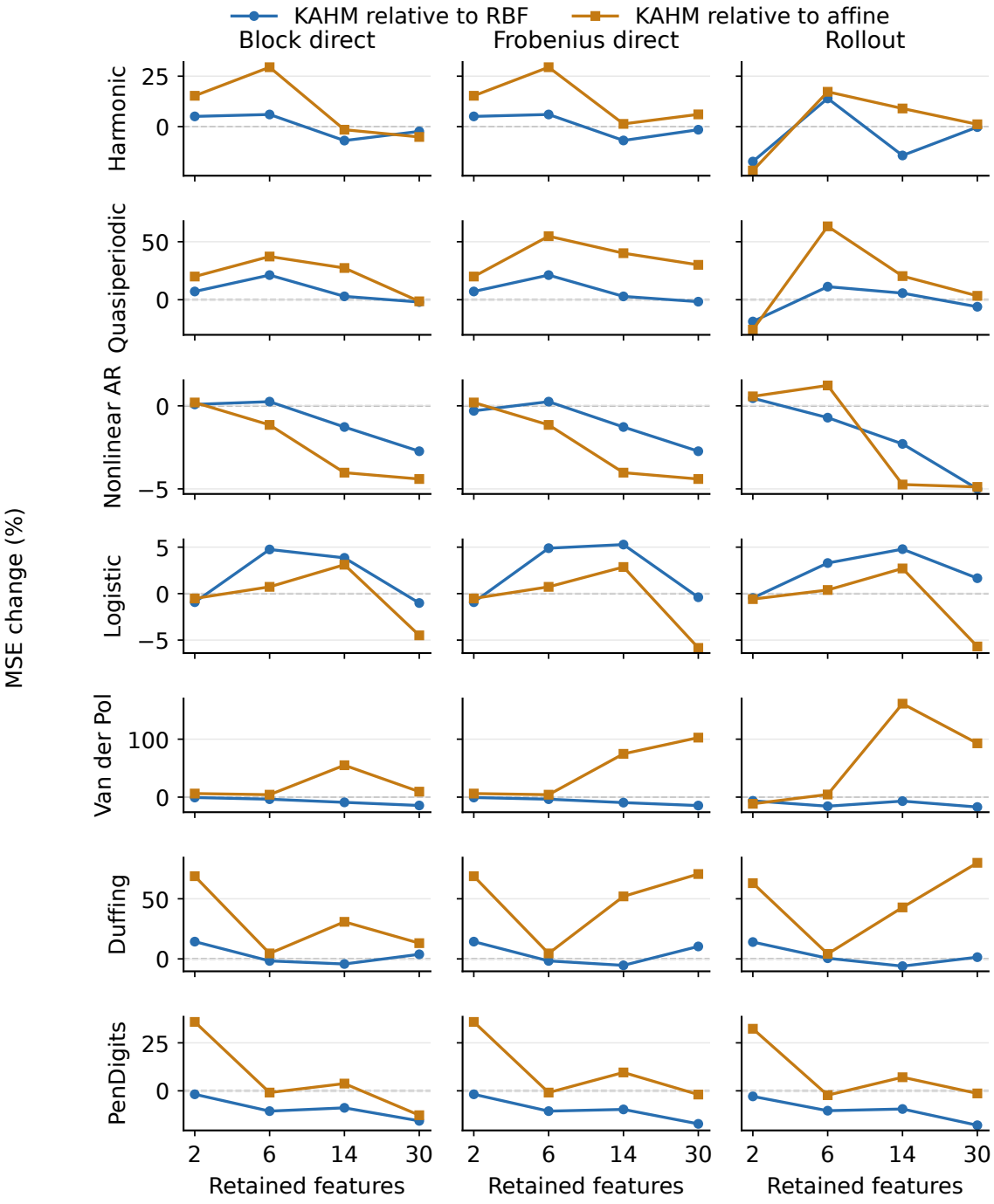


Figure 5: Percentage change in mean all-horizon test MSE at each fixed retained dimension: $100(\mathrm{MSE}_{\mathrm{KAHM}}/\mathrm{MSE}_{\mathrm{alternative}} - 1)$, where each MSE is averaged over five fitting seeds. Negative values favor KAHM. History length and preprocessing are selected by validation at each point. PenDigits first averages coordinate losses and weights writers equally.

8 Discussion and limitations

Contribution to KAHM forecasting. The procedure combines KAHM reconstruction features with a common observed prediction target, a constrained decoder class, and calibrated accuracy decisions. Retaining feature blocks preserves earlier predictors as the representation grows. The matched study finds consistent KAHM gains on the nonlinear autoregressive task and against RBF geometry on the fixed PenDigits cohort. Affine-hull geometry with Frobenius-constrained or rollout decoders is stronger on the oscillators. The usefulness of KAHM geometry therefore depends on the task, decoder, and retained resolution. The reconstruction algorithm is inherited; the certification theory applies to each representation under its stated assumptions.

Three distinct uses of calibration. Validation error ranks fitted candidates. Calibration adds an upper confidence bound that can certify a population error tolerance, a lower bound that can exclude that tolerance for an entire sensitivity-constrained decoder class, and a paired comparison that can certify improvement between two fitted models. Each answers a different question. Acceptance supports the requested mean squared error under the evaluation law. Exclusion can motivate changing the representation or budget. A gap between the bounds can reflect uncertainty or a loose population obstruction and need not identify a defect in fitting.

Limits of refinement guarantees. The method learns reconstruction models and constrained decoders, then selects among retained dictionary prefixes. It does not prove that geometric clustering discovers a sufficient state. Residual correlation with a new feature block, together with unused coefficient budget, gives a sufficient population condition for one improving direction. Distance expansion gives a necessary condition for meeting a target accuracy. Neither result guarantees that an arbitrary split will help. The experiments evaluate fixed candidate families; sequential refinement with fresh evaluation blocks is a theoretical extension that was not tested here.

Restricted classes and loose lower bounds. Excluding all κ-Lipschitz decoders of one representation does not exclude predictors with greater sensitivity, longer histories, or additional observations. The pairwise obstruction can also be zero when the best risk in the class is positive. Pairing only at a common origin, for example, can miss information loss that occurs across origins. A different, class-conditioned pairing law may yield a more informative bound, but requires its own sampling protocol. Zero obstruction is therefore neither a certificate of adequacy nor a feature learning objective in this study.

Independent units and statistical precision. Independent episodes are appropriate for independently initialized simulations and can be appropriate for separate experimental runs. A chronological split of one long record does not itself supply independent episodes. Extending the certificate to a dependent series requires quantitative assumptions and a justified correction for dependence both within calibration and between training and calibration. An informal autocorrelation estimate or chosen "effective sample size" does not supply that correction. The PenDigits study illustrates another limitation: many strokes and time points can still provide few defensible independent writer-level units. With so few units, valid conservative bounds may fail to certify a useful tolerance.

Target bounds and distribution shift. The synthetic output ranges follow from the task definitions. PenDigits uses the acquisition box and a fixed linear scale, without normalizing a completed trajectory by its future extrema. For unbounded targets, the present bounded-loss certificate needs a different tail assumption or an explicitly capped loss; clipping predictions alone does not bound the error.

Calibration under one episode law does not certify performance after changes in dynamics, noise, acquisition, or origin selection. These requirements apply to all comparison methods.

Preprocessing as a model choice. Training-based range scaling can improve retained soft KAHM forecasts, including across repeated fits on the fixed PenDigits cohort (Section 6). The same study finds regressions on other tasks and with hard assignments. Preprocessing should therefore be an explicit, validated choice. These results do not justify replacing standardization throughout the original experiments or assuming that bounded inputs improve the geometry. Certification after choosing among preprocessing options must also account for the enlarged family of candidate statements.

Follow-up design and calibration reuse. The original-seed preprocessing comparison reuses calibration episodes after earlier results motivated the follow-up. Its computed endpoints are therefore nominal summaries; fixing the rerun configuration does not establish independence from previously observed calibration results. The synthetic confirmation and matched-geometry experiments use new calibration episodes. Their confidence statements apply within each declared task and fitting repetition, under the specified allocation of failure probability. PenDigits follow-ups instead reuse the previously examined writer cohort. These comparisons are descriptive; an independent-writer sampling assumption alone does not justify confidence claims after adapting the study to that cohort.

Experimental and computational scope. The study covers controlled dynamics and one real acquisition domain. The matched comparisons use specified implementations and shared selection grids; separate tuning for each family could change their ranking. They do not establish an accuracy frontier against broadly tuned neural forecasters. Feature dimension measures representation size, not runtime. The prototype recomputes some shared transforms, so its candidate-level training times do not provide a fair speed comparison. Archived execution times and environments document the experiments rather than establish deployment-speed advantages.

9 Conclusion

KAHM soft associations can support forecasts of the observed signal and finite-sample statements about their expected squared error. Retained dictionaries with a block coefficient budget preserve earlier predictors and prevent duplication from artificially enlarging the class. The pairwise obstruction, fitted-risk upper bound, and paired improvement bound distinguish representation limits from the performance of learned forecasts. The analysis also supports comparisons with other geometric dictionaries.

The experiments show favorable KAHM comparisons on nonlinear autoregression and pen trajectories, alongside stronger oscillator alternatives, preprocessing regressions, and unresolved accuracy decisions. Informative certification requires credible target bounds, an appropriate evaluation law, and enough independent observations at the episode level.

Code, data, and research transparency

The accompanying repository provides the implementation, frozen configurations, result summaries, and figure generators in `extensions/certified_forecasting`; editable manuscript files are in its `paper` directory. The original experiments, input-scaling follow-up, and matched geometry–decoder comparison are archived separately. Their records include configurations, source hashes, and

documented exact replays of damaged score archives in the scaling study. The evidence packages identify retained score arrays, and fixed-seed runners regenerate the full episode-level archives. Appendix B specifies the experimental configurations and independent sampling units.

Use of AI tools. AI tools assisted with derivation exploration, implementation, literature retrieval, and manuscript preparation.

Appendix A. Convex fitting and a numerical optimality certificate

This appendix derives the projection used to fit the decoder and a computable upper bound on its training-objective suboptimality. That optimization bound is distinct from the population prediction-risk certificate: one measures how accurately a finite training problem was solved, and the other evaluates forecasting error on new episodes.

Let N be the number of training windows and $p_s = \sum_{b=1}^{s} C_b$ the total number of retained feature coordinates. Write $F = [F_1, \dots, F_s] \in \mathbb{R}^{N \times p_s}$ for the fixed feature matrix, where $F_b \in \mathbb{R}^{N \times C_b}$ contains block b. Let $\mathsf{Y} \in \mathbb{R}^{N \times q}$ contain the corresponding stacked targets, with $q = d_y H$ for d_y output coordinates at each of H horizons. Training windows may be dependent; optimization of this finite objective requires no independence assumption on its rows.

With intercept $a \in \mathbb{R}^q$ and coefficient matrix $W \in \mathbb{R}^{p_s \times q}$, solve

$$\min_{W,a} \frac{1}{N} \|\mathsf{Y} - \mathbf{1}a^\top - FW\|_{\mathrm{F}}^2 + \lambda \|W\|_{\mathrm{F}}^2 \quad \text{subject to} \quad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} \le \kappa. \tag{20}$$

Here $\mathbf{1} \in \mathbb{R}^N$ is the all-ones column vector, $W_b \in \mathbb{R}^{C_b \times q}$ contains the rows associated with F_b, $\lambda \ge 0$ is the ridge coefficient, and $\kappa \ge 0$ is the block-norm budget. The intercept is neither penalized nor constrained. The loss sums over output coordinates and horizons, then averages over training windows; there is no additional division by H or q. Clipping is applied after fitting. Thus (20) minimizes a convex objective before clipping, not the evaluated clipped empirical loss.

Define the row means $\bar{F} = N^{-1}\mathbf{1}^\top F$ and $\bar{\mathsf{Y}} = N^{-1}\mathbf{1}^\top \mathsf{Y}$, and the centered matrices

$$F_c = F - \mathbf{1}\bar{F}, \qquad \mathsf{Y}_c = \mathsf{Y} - \mathbf{1}\bar{\mathsf{Y}}.$$

For fixed W, minimizing over the intercept gives $a^\top = \bar{\mathsf{Y}} - \bar{F}W$. Substituting this expression eliminates the intercept. With I_{p_s} the $p_s \times p_s$ identity matrix, put

$$M = F_c^\top F_c/N + \lambda I_{p_s} \in \mathbb{R}^{p_s \times p_s}, \qquad C = F_c^\top \mathsf{Y}_c/N \in \mathbb{R}^{p_s \times q}, \qquad c = \|\mathsf{Y}_c\|_{\mathrm{F}}^2/N.$$

The reduced objective and its gradient are

$$J(W) = \operatorname{tr}(W^\top M W) - 2\langle W, C\rangle_{\mathrm{F}} + c, \qquad G(W) = \nabla J(W) = 2(MW - C).$$

The Frobenius inner product is $\langle A, B\rangle_{\mathrm{F}} = \operatorname{tr}(A^\top B)$. Since M is positive semidefinite, J is convex. Moreover,

$$\|G(W) - G(V)\|_{\mathrm{F}} \le 2\|M\|_{\mathrm{op}} \|W - V\|_{\mathrm{F}},$$

where $\|M\|_{\mathrm{op}} = \sup_{\|v\|_2 = 1} \|Mv\|_2$ is the operator norm. The implementation uses projected gradient steps with acceleration and monotone restarts. The following formulas assess feasibility and optimization accuracy without requiring a convergence-rate claim for that implementation.

Proposition 8 (Projection onto the block norm ball) *Let $\mathcal{C}_\kappa = \{W : \sum_b \|W_b\|_{\mathrm{F}} \le \kappa\}$ be the feasible coefficient set, and let $A \in \mathbb{R}^{p_s \times q}$ have the same row-block partition as W. Define $a_b = \|A_b\|_{\mathrm{F}}$ and let $\mathcal{P}_\kappa(A)$ denote the nearest matrix in $\mathcal{C}_\kappa$ in Frobenius norm. If $\sum_b a_b \le \kappa$, then $\mathcal{P}_\kappa(A) = A$. Otherwise choose a threshold $\theta \ge 0$ satisfying $\sum_b (a_b - \theta)_+ = \kappa$. The projection is*

$$\big[\mathcal{P}_\kappa(A)\big]_b = \begin{cases} (1 - \theta/a_b)_+ A_b, & a_b > 0, \\ 0, & a_b = 0. \end{cases} \tag{21}$$

In particular, $\mathcal{P}_0(A) = 0$. For every feasible W, the objective gap satisfies

$$0 \le J(W) - J^* \le \mathcal{D}(W) := \langle G(W), W \rangle_{\mathrm{F}} + \kappa \max_b \|G_b(W)\|_{\mathrm{F}}, \tag{22}$$

where $J^* = \min_{V \in \mathcal{C}_\kappa} J(V)$ is also the optimum value of (20)*, and $G_b(W)$ is the row block of the gradient corresponding to W_b.*

Proof For a block B_b constrained to have Frobenius norm $v_b \ge 0$, Cauchy–Schwarz gives

$$\|B_b - A_b\|_{\mathrm{F}}^2 = v_b^2 + a_b^2 - 2\langle B_b, A_b \rangle_{\mathrm{F}} \ge (v_b - a_b)^2.$$

When $a_b > 0$, equality is attained by taking $B_b = (v_b/a_b)A_b$. Thus matrix projection reduces to the scalar problem

$$\min_{v_1, \ldots, v_s \ge 0} \sum_{b=1}^{s} (v_b - a_b)^2 \quad \text{subject to} \quad \sum_{b=1}^{s} v_b \le \kappa.$$

If the original norms are feasible, the solution is $v_b = a_b$. Otherwise the constraint is active, and the Karush–Kuhn–Tucker optimality conditions give $v_b = (a_b - \theta)_+$, with θ chosen to make the sum equal to κ. Rescaling each nonzero block gives (21); a zero block remains zero.

For the objective gap, convexity gives, for every feasible V,

$$J(W) - J(V) \le \langle G(W), W - V \rangle_{\mathrm{F}}.$$

Also,

$$\langle G(W), V \rangle_{\mathrm{F}} \ge -\sum_b \|G_b(W)\|_{\mathrm{F}} \|V_b\|_{\mathrm{F}} \ge -\kappa \max_b \|G_b(W)\|_{\mathrm{F}}.$$

The last value is attainable by assigning the entire budget to a block with largest gradient norm, in the direction opposite to that gradient (or by $V = 0$ if the gradient is zero). Substitute an optimal V in the convexity inequality to obtain (22). ■

The implementation records coefficient norms, the gap $\mathcal{D}(W)$, iteration count, and whether the requested gap tolerance was reached. These are floating-point diagnostics, not interval-arithmetic guarantees. The held-out risk certificate requires the returned model to satisfy the Lipschitz budget used in calibration; it does not require an optimal training solution.

For a joint multihorizon fit, let $W_{b,h}$ be the columns of block W_b associated with the output at horizon h. Then

$$\sum_b \|W_{b,h}\|_{\mathrm{F}} \le \sum_b \|W_b\|_{\mathrm{F}} \le \kappa,$$

so the joint budget also bounds each horizon decoder's sensitivity. For scalar outputs, $W_{b,h}$ is simply column h and its Frobenius norm is its Euclidean norm. Although the block constraint is invariant to exact feature duplication, the ridge term $\lambda \|W\|_{\mathrm{F}}^2$ is not: splitting coefficients among duplicate blocks changes that penalty. The duplicate-block optimization test therefore uses $\lambda = 0$.

Appendix B. Synthetic protocols, comparison models, and reproduction

B.1 Episode laws

Each row of a generated data array is one independent episode. In the following definitions, t is an integer time index, y_t is the observed scalar signal, and $\mathrm{Unif}[a, b]$ denotes the uniform distribution on $[a, b]$. All random draws are independent unless an episode-level parameter is explicitly shared across times. The observation range is contained in $[-1, 1]$ for all four main systems. Clipping forecasts to this interval gives scalar squared losses in $[0, 4]$.

1. **Harmonic.** Draw one phase $\theta \sim \mathrm{Unif}[0, 2\pi]$ and one angular frequency $\nu \sim \mathrm{Unif}[.12, .28]$ per episode. Define
$$y_t = .85\cos(\theta + \nu t) + \eta_t, \qquad \eta_t \sim \mathrm{Unif}[-.03, .03],$$
where η_t is observation noise. Thus $|y_t| \le .88$.

2. **Quasiperiodic.** Draw phases θ_1 and θ_2 independently from $\mathrm{Unif}[0, 2\pi]$, and draw $\nu \sim \mathrm{Unif}[.10, .22]$ once per episode. Set
$$y_t = .55\cos(\theta_1 + \nu t) + .30\cos(\theta_2 + \sqrt{2}\nu t) + \eta_t, \qquad \eta_t \sim \mathrm{Unif}[-.03, .03].$$
The two angular frequencies are ν and $\sqrt{2}\nu$. Again $|y_t| \le .88$.

3. **Nonlinear autoregression.** Initialize y_0 and y_1 independently from $\mathrm{Unif}[-.8, .8]$. For $t \ge 1$, apply
$$y_{t+1} = \tanh\{1.25y_t - .65y_{t-1} + .25\sin(3y_t) + \eta_{t+1}\}, \qquad \eta_{t+1} \sim \mathrm{Unif}[-.10, .10],$$
where η_{t+1} is an innovation and $\tanh$ denotes the hyperbolic tangent. The implementation discards the first 80 stored values before retaining a length-72 episode. The two initial values occupy indices zero and one; recursive updates start at index two. The hyperbolic tangent ensures the stated output bound.

4. **Logistic.** Draw the episode parameter $r \sim \mathrm{Unif}[3.65, 3.95]$ and initial internal state $u_0 \sim \mathrm{Unif}[.05, .95]$, then iterate $u_{t+1} = ru_t(1 - u_t)$. After 40 burn-in updates, store the next 72 updated values, transformed to the observed scale by $y = 2u - 1$. Because $r \le 4$, updates preserve $u \in [0, 1]$ and hence $y \in [-1, 1]$. Within-episode dynamics can be chaotic. Independence between episodes follows from independent initialization, without requiring a mixing argument.

The finite generation and origin protocol defines the risk being estimated. No assumption of stationarity after burn-in or sampling from an invariant distribution is needed. Each generator has its own observation scale and horizon interpretation; we do not average errors across generators as if they were a common physical measure.

B.2 Random streams and configuration

The main fitting seeds are $1701, \ldots, 1705$. For a numerical seed s, the four NumPy generator seeds are $100s + 11$ for training, $100s + 23$ for validation, $100s + 47$ for calibration, and $100s + 89$ for testing. These specify separate pseudorandom streams. Reusing the same seed conventions across generators supports reproducible comparisons; a within-run confidence guarantee does not require

independence across benchmark conditions. Clustering receives an offset determined by the training seed and history length, and each retained block receives its own deterministic offset. The archived environment records the library versions and default bit generators used for all random draws.

Geometry construction can use a fixed seeded subset of at most 1600 training rows. The decoder uses all 1664 rows, obtained from 128 episodes and 13 origins per episode. These rows may overlap: the finite optimization objective does not require independent training rows. Validation and test losses weight episodes equally and origins equally within each episode. All models use the same origin sets, and validation selection averages MSE over all three horizons.

B.3 Comparison definitions

The RBF and affine-hull comparators use the same type of k-means partition, with five initializations and at most 150 iterations. Here k is the number of coordinates in the block. To define their scores, suppress the block index. Let $\xi \in \mathbb{R}^{d_x}$ be a preprocessed history, where d_x is its input dimension. For cluster $c \in \{1, \dots, k\}$, let $\mu_c \in \mathbb{R}^{d_x}$ be its fitted center, $\mathcal{I}_c$ the nonempty set of assigned geometry-training indices, and $n_c = |\mathcal{I}_c|$. Define the cluster's squared-radius scale and the global RBF scale by

$$s_c = \max\left\{\frac{1}{n_c}\sum_{i\in\mathcal{I}_c}\|\xi_i - \mu_c\|_2^2,\ 10^{-6}\right\}, \qquad \beta = \mathrm{median}_{1\le c\le k}\, s_c,$$

where median is the usual sample median. The RBF score is

$$a_c^{\mathrm{RBF}}(\xi) = \frac{\|\xi - \mu_c\|_2^2}{2\beta}.$$

Thus every cluster uses the same bandwidth, while the positive floor prevents division by a zero radius.

For the affine-hull comparator, take the singular value decomposition of the matrix whose rows are $(\xi_i - \mu_c)^\top$, $i \in \mathcal{I}_c$. When $n_c \ge 2$, let V_c contain its first $k_c = \min\{2, d_x - 1, n_c\}$ right singular vectors as columns. For a singleton cluster, set $k_c = 0$ and use an empty basis. Then $P_c = V_c V_c^\top$ is the orthogonal projection onto the selected local subspace; $P_c = 0$ for an empty basis. The constraint $k_c \le d_x - 1$ leaves an orthogonal direction whenever one is available. Define

$$r_c = \max\left\{\frac{1}{n_c}\sum_{i\in\mathcal{I}_c}\|(I_{d_x} - P_c)(\xi_i - \mu_c)\|_2^2,\ 0.05 s_c,\ 10^{-6}\right\},$$

$$a_c^{\mathrm{AH}}(\xi) = \frac{\|(I_{d_x} - P_c)(\xi - \mu_c)\|_2^2}{r_c} + 0.1\frac{\|\xi - \mu_c\|_2^2}{s_c},$$

where I_{d_x} is the identity matrix. The first term measures distance from the local affine subspace; the second penalizes distance from its center. The floors in r_c keep the normalization positive even when the training residuals vanish.

For either choice $a_c = a_c^{\mathrm{RBF}}$ or $a_c = a_c^{\mathrm{AH}}$, the block coordinates are

$$\Phi_c(\xi) = \frac{\exp(-a_c(\xi))}{\sum_{j=1}^k \exp(-a_j(\xi))}.$$

The implementation evaluates this normalization in log space. Its optional score multiplier is fixed at one in the reported experiments. These formulas and numerical conventions are implemented in

`src/certified_forecasting/features.py` and are unchanged from the original code baseline `a8bcf6c8016499e1a4b64f7dd4196af43c60bd78`. Both geometric alternatives use the same retained-block decoder.

Hard KAHM states replace the final block by a one-hot vector: its largest coordinate becomes one and all others become zero. Ridge AR uses the original causal history. Quadratic regression uses training-standardized histories and all monomials of degree one or two, including cross-products; it does not add a constant feature because the intercept is unpenalized. Both direct decoders use the stated Frobenius constraint and ridge coefficient. Persistence repeats the latest observation at every horizon. The constant model predicts the training target mean separately for each horizon and has sensitivity budget zero.

For the iterated baselines, let $z_t \in \mathbb{R}^{1\times p}$ be the row vector of p features at time t. Ridge regression fits an affine feature transition and a scalar observation decoder,

$$z_{t+1} \approx z_t M + b, \qquad y_t \approx z_t v + c,$$

where $M \in \mathbb{R}^{p\times p}$ is the transition matrix, $b \in \mathbb{R}^{1\times p}$ is its intercept, $v \in \mathbb{R}^p$ is the observation coefficient vector, and $c \in \mathbb{R}$ is the observation intercept. For a positive integer horizon h, composing these fitted maps gives the un-clipped forecast

$$\widetilde{y}_{t+h} = z_t M^h v + b\left(\sum_{\ell=0}^{h-1} M^\ell\right) v + c,$$

with M^0 the identity matrix. Only this final scalar forecast is clipped. The effective output coefficient is $M^h v$; its Euclidean norm is the frozen Lipschitz budget relative to Euclidean feature distance at horizon h. The certificate evaluates this fitted forecast directly and requires no spectral-stability assumption, exact feature closure, or approximation to the trajectory distribution.

B.4 Reproducing the retained results

The code release provides the main configurations, source hashes, per-repetition candidate descriptions, certificate endpoints, scalar trial records, and generated tables. Double-precision per-episode score archives allow independent reconstruction of each confidence calculation. The full evidence package retains the original archives; fixed-seed runners regenerate any large archives omitted from the code repository. A deterministic replay was checked for one complete main experiment, and a score audit covers the entire retained main campaign.

From the extension directory, run

```
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1
export PYTHONPATH=src
python experiments/run_forecasting.py \
  --config configs/main_v1.json --output results/reproduced_main
python experiments/run_known_risk.py \
  --output results/reproduced_known_risk
python experiments/prepare_pendigits.py
python experiments/run_pendigits.py --help
python -m unittest discover -s tests -v
```

The documentation supplies exact commands for comparing reproduced and retained results, running the real-data and follow-up studies, generating figures, and building the manuscript. Reference package versions are NumPy 2.3.5, SciPy 1.17.0, scikit-learn 1.8.0, Matplotlib 3.10.8, pandas 2.2.3, and joblib 1.5.3. Bitwise reproducibility across numerical libraries or architectures is not assumed. The runner refuses to overwrite an existing run and records exceptions instead of silently omitting failed attempts.

B.5 Reproducing the input-scaling follow-up

The original numerical campaigns remain the reference experiments. The separate input-scaling study retains its original-seed and confirmation analyses in `results/input_scaling_study/analysis_original` and `results/input_scaling_study/analysis_confirmation`, respectively. Its study directory contains frozen protocols, per-seed comparisons, source provenance, and score-archive repair records. From the extension directory, run the full four-arm schedule and generate its manuscript tables with

```
python experiments/prepare_pendigits.py
python experiments/run_input_scaling.py \
  --output results/reproduced_input_scaling --workers 4
python scripts/make_scaling_assets.py
```

The runner uses a new output directory and freezes both phase schedules before execution. Its `--phases`, `--campaigns`, and `--modes` options restrict a reproduction to specified parts of the study. By default, the asset generator reads the released audited comparison CSVs; alternative analysis directories can be supplied through `--analysis` and `--confirmation`. It does not refit models or select preprocessing arms. The original code baseline is commit `a8bcf6c8016499e1a4b64f7dd4196af43c60bd78`. The release retains the original source hashes, so historical results remain distinguishable from outputs of the new optional-scaling interface.

B.6 Reproducing the matched-geometry follow-up

The new configurations enable matched combinations of geometric representations and decoders, with per-geometry input preprocessing; historical configurations retain their original candidate families. Each scalar fit in one preprocessing arm contains 72 matched candidates and 12 additional controls. The refinement allocation covers 54 adjacent-prefix horizon comparisons: three geometries, two history lengths, three adjacent prefix pairs, and three horizons. The complete schedule and source/configuration hashes are frozen before the full fits. Section 7 and the retained protocol specify the selection rules and joint confidence allocation.

The primary, fixed-dimension, and validation-selected-head summaries are in `taskwise_comparisons.csv`. The retained analysis directory is `results/matched_geometry_study/analysis`.

With PenDigits prepared as above, the reproduction sequence freezes the schedule and then resumes fitting with `experiments/run_matched_geometry.py`. It aggregates the results with `experiments/analyze_matched_geometry.py` and creates manuscript assets with `scripts/make_matched_assets.py`. Complete commands are provided in the source package's `README_BUILD.txt` and the repository reproduction guide. The analysis checks completeness and score-archive integrity before aggregating paired comparisons. It retains every selected

	Block direct			Frobenius direct			Rollout		
Task	KAHM	RBF	Affine	KAHM	RBF	Affine	KAHM	RBF	Affine
Harmonic	0.04977	0.05097	0.05245	0.04767	0.04841	0.04494	0.05226	0.05238	0.05166
Quasiperiodic	0.03426	0.03495	0.03474	0.03372	0.03430	0.02592	0.03922	0.04175	0.03795
Nonlinear AR	0.00456	0.00469	0.00477	0.00456	0.00469	0.00477	0.00457	0.00481	0.00481
Logistic	0.16548	0.16718	0.17327	0.16066	0.16129	0.17063	0.16763	0.16490	0.17774
Van der Pol	0.05292	0.06182	0.04835	0.02642	0.03092	0.01303	0.03336	0.04023	0.01730
Duffing	0.02482	0.02391	0.02195	0.02341	0.02121	0.01372	0.02741	0.02702	0.01523
PenDigits	0.00514	0.00609	0.00589	0.00504	0.00609	0.00514	0.00546	0.00665	0.00554

Table 14: Absolute test MSE for all nine matched geometry–head families in the follow-up. Each cell averages the predefined horizons and five fitting seeds after validation-only selection of preprocessing, history and retained dimension within that family. PenDigits additionally pools both selected coordinate losses with equal eligible-writer weights on the same evaluation cohort. Corresponding seed variation, per-horizon losses and exact selected candidates are retained in the repository.

candidate, per-seed effect, per-horizon result, and PenDigits writer comparison. The asset generator reads the complete analysis without fitting or selection. Source snapshots and scalar records are stored separately from historical results. The earlier KAHM preprocessing comparison remains a distinct experiment from the matched comparison of all geometric families.

Table 14 gives the absolute MSEs corresponding to the relative effects in the main results, including every matched geometry–decoder family. The complete study retains 13,440 candidates. Source, configuration, array-integrity, and independently reconstructed score checks are recorded with the analysis.

Appendix C. Exact-risk benchmark and centered comparator

This benchmark has an analytically known optimal risk and fitted-model risk, so certificate coverage can be checked without using a test estimate as population truth. Let $Z \in \{0, 1\}$ be a Bernoulli state with $\Pr(Z = 1) = p \in (0, 1)$, and let the scalar target be $Y = 2Z-1$. For a representation-separation parameter $a \in [0, 1]$, use

$$\Phi_a(0) = ((1 + a)/2, (1 - a)/2)\,, \qquad \Phi_a(1) = ((1 - a)/2, (1 + a)/2)\,.$$

Both vectors lie in the two-coordinate probability simplex. Their Euclidean distance is $\sqrt{2}a$, whereas the target separation is 2. For sensitivity budget $\kappa \geq 0$, define the unexplained separation $c = (2 - \kappa\sqrt{2}a)_+$. Proposition 2 then gives

$$R^* = p(1 - p)c^2.$$

Here R^* denotes the minimum MSE over the benchmark's constrained predictors. The two target values, representation family, and support are specified in advance by the benchmark design.

An independent training sample gives the empirical frequency $\widehat{p}$ of state 1. We choose the constrained empirical-risk minimizer whose predictions at the two states are

$$\widehat{f}(0) = -1 + \widehat{p}c, \qquad \widehat{f}(1) = 1 - (1 - \widehat{p})c. \tag{23}$$

These values lie in $[-1,1]$ and have separation $2-c=\min\{2,\kappa\sqrt{2}a\}$. The affine construction in Proposition 2 therefore realizes them within the budget; when $a=0$, the two values coincide. If one state is absent from training, (23) selects one feasible empirical minimizer using the predeclared support. This fixes an otherwise ambiguous prediction at the unobserved state.

Conditional on the training sample, state 0 has prediction error $-\widehat{p}c$, and state 1 has error $(1-\widehat{p})c$. Consequently the exact fitted population risk is

$$R(\widehat{f})=c^2\{(1-p)\widehat{p}^2+p(1-\widehat{p})^2\}=R^*+c^2(\widehat{p}-p)^2. \tag{24}$$

For an independent pair of calibration states, the witness score equals $c^2/2$ when the states differ and zero otherwise. The mismatch probability is $2p(1-p)$, so the expected score is exactly R^*. With n calibration observations, the pair lower bound uses $\lfloor n/2\rfloor$ disjoint pairs. An odd final observation is excluded from pairing but remains in the fitted-risk and centered statistics. The reported calibration sizes are all even.

The centered comparator. For comparison, use the raw-target Euclidean version of the earlier centered certificate with one reference class. Let $Z_1,\dots,Z_n$ be independent calibration states, with $n\ge 4$, and $\bar{z}=n^{-1}\sum_{i=1}^n Z_i$ their empirical frequency. The output and representation variances, computed with denominator n, are

$$\widehat{s}=4\bar{z}(1-\bar{z}),\qquad \widehat{t}=2a^2\bar{z}(1-\bar{z}).$$

Their population counterparts are $\sigma^2=4p(1-p)$ and $\tau^2=2a^2p(1-p)$. The population centered bound is $(\sigma-\kappa\tau)_+^2\le R^*$, using nonnegative square roots. In this two-state benchmark the inequality is an equality.

For a failure probability $\eta\in(0,1)$, define

$$\begin{aligned} r&=\log(2/\eta)/n,\\ V_\eta&=(\widehat{s}-4\sqrt{r/2})_+,\\ T_\eta&=\min\{1,\widehat{t}+\sqrt{2r}+1/n\},\\ L_\eta^{\mathrm{ctr}}&=(\sqrt{V_\eta}-\kappa\sqrt{T_\eta})_+^2. \end{aligned} \tag{25}$$

Here V_η is a lower bound on σ^2, T_η is an upper bound on τ^2, and the superscript "ctr" identifies the centered risk bound. We show that $L_\eta^{\mathrm{ctr}}\le R^*$ with probability at least $1-\eta$.

First, $\mathbb{E}\widehat{s}=(1-1/n)\sigma^2\le\sigma^2$, and changing one calibration observation changes $\widehat{s}$ by at most $4/n$. The bounded-differences inequality therefore gives

$$\widehat{s}-\mathbb{E}\widehat{s}\le 4\sqrt{r/2}$$

with failure probability at most $\eta/2$. On that event, $V_\eta\le\sigma^2$. Second, changing one observation changes $\widehat{t}$ by at most $2/n$, so the same inequality gives

$$\mathbb{E}\widehat{t}-\widehat{t}\le\sqrt{2r}$$

with failure probability at most $\eta/2$. Since $\mathbb{E}\widehat{t}=(1-1/n)\tau^2$ and $\tau^2\le 1$,

$$\tau^2\le\widehat{t}+\sqrt{2r}+\tau^2/n\le\widehat{t}+\sqrt{2r}+1/n.$$

Capping the right side at one remains valid because $\tau^2 \leq 1$. A union bound now gives $V_\eta \leq \sigma^2$ and $T_\eta \geq \tau^2$ jointly with probability at least $1 - \eta$. The expression in (25) increases with its first variance argument and decreases with its second. Hence it is no larger than $(\sigma - \kappa\tau)_+^2 \leq R^*$ on that event. The constant one is the general simplex-variance bound; it is conservative for this two-coordinate example.

In each Monte Carlo trial, the fitted-risk upper bound, pair-witness lower bound, and centered lower bound each receive failure probability $\delta/3$. For the centered bound, setting $\eta = \delta/3$ splits that allocation between its two internal variance bounds. The three reported directions therefore hold jointly with probability at least $1 - \delta$ within a trial. No joint $1 - \delta$ event is asserted across all trials, grid conditions, or separately reported experiments.

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